

Small-scale Database implementation - Tables and relationships

Week 5

Dr. Farshid Keivanian

Week 5 Outline

1. Why use Access instead of Excel for databases?
2. Introduction to MS Access
3. Windows Requirement
4. If You Want to Install Microsoft Access On You Windows Laptop
5. Run Microsoft Access
6. Menus in Microsoft Access
7. MS Access Objects
8. MS Access Environment (Ribbons & Navigation Pane)
9. Tables in Access
10. Data Types in Access
11. Activity 1: Building a Table and Adding Rules to Fields
12. Activity 2: Mini-System: City & Student
13. Activity 3: Library Books & Loans
14. Relationships in Access
15. Tutorial Week 5 Preparation
16. Role-Playing Debate
17. Review and Feedback on Tutorial Week 4

Why use Access instead of Excel for databases?

Access enforces structure (data types, relationships, referential integrity), reduces errors, supports multi-table queries/forms/reports, and scales better for relational tasks than a single spreadsheet.

MS Access is a small-scale relational DBMS. It lets us store, manage, and query data using tables, queries, forms, reports, and relationships.

Analogy:

Think of it as a digital filing cabinet. Each drawer = a table, folders = records, and labels = fields.

Microsoft Access is a Windows-based application, so you must use a Windows laptop or a Windows PC to run it. It is designed specifically for Microsoft Windows operating systems such as Windows 10 and Windows 11

In our lab, all PCs already have Windows installed along with Microsoft Access, so you can directly use them without any additional setup.

If you are using a Mac, Microsoft Access is not natively supported. You will need to install a virtual machine (VM) to run Windows on your Mac. A virtual machine allows you to use Windows inside macOS, so you can install and run Access just like on a normal Windows computer.

One commonly used option is:

- Parallels Desktop (many students use it to run Windows on Mac)

<https://www.parallels.com/?srsltid=AfmBOoreRhN8t1kxArDr1GZlzl1WMT3CctxNQzUVrik61xDmR-3lSKlw>

Before choosing a VM, make sure your Mac has enough resources (RAM, storage, and CPU) because running Windows virtually requires additional system power.

You may explore that link and research this option. While I have not personally used it, some students have found it useful. If it works well for you, you may continue using this VM on your Mac.

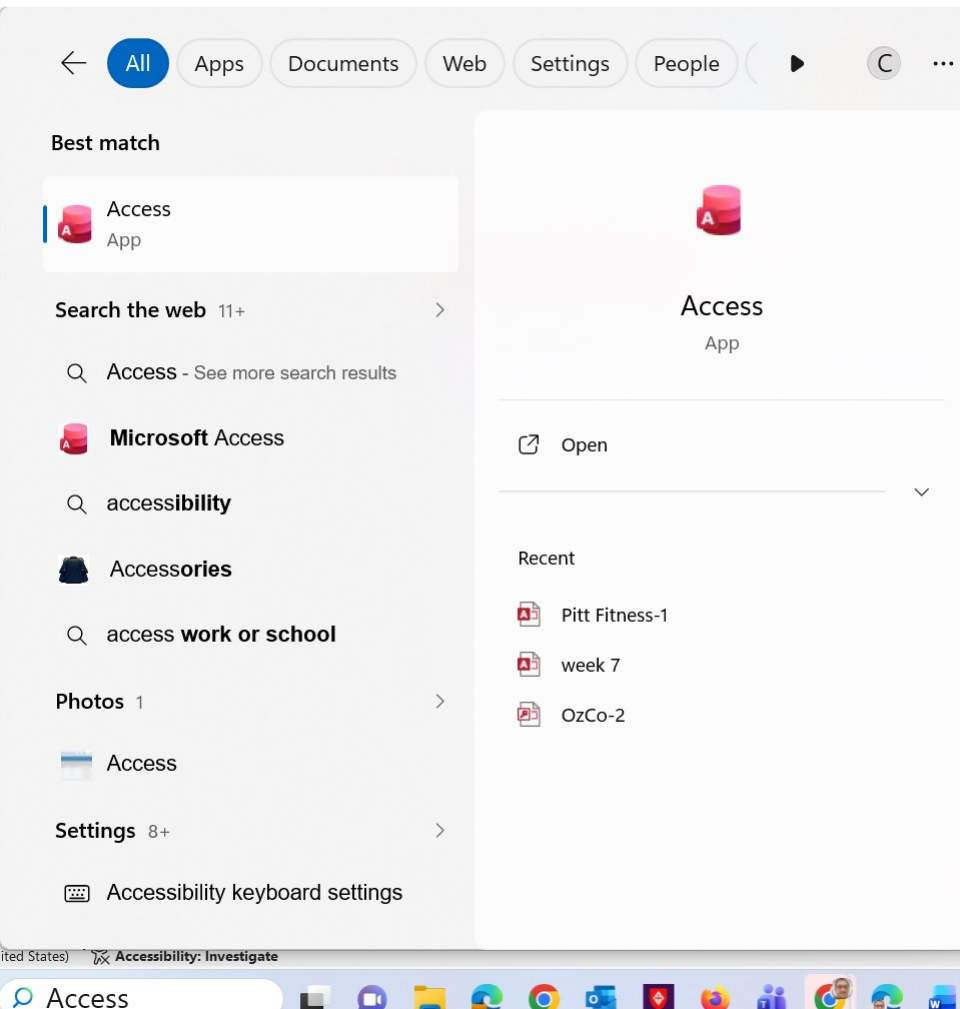
In summary:

- Best option: Use lab PCs (already set up)
- Alternative: Use a Windows laptop
- Mac users: Install a VM (e.g., Parallels) to run Windows and Access

Lab PC - Microsoft Access

Use Library or Lab PCs to access Microsoft Access

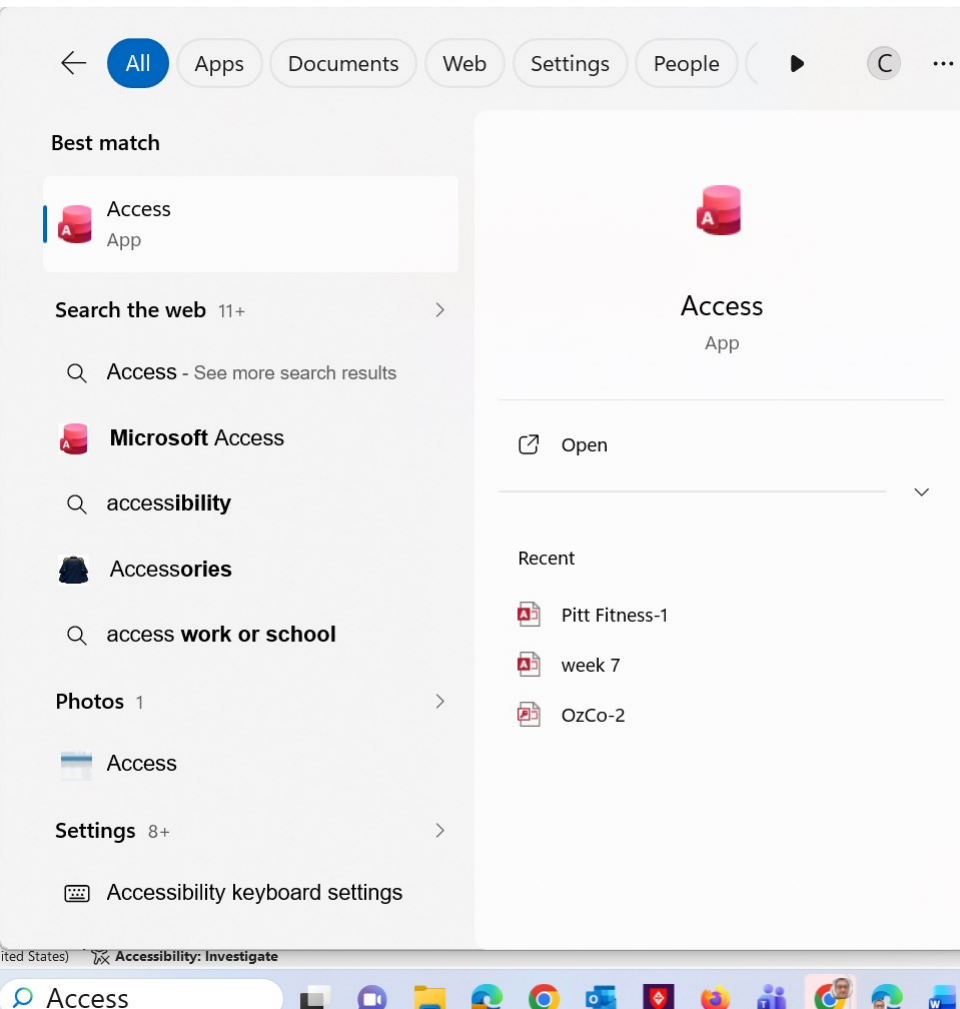
In Windows >> Search for Access and Open the App



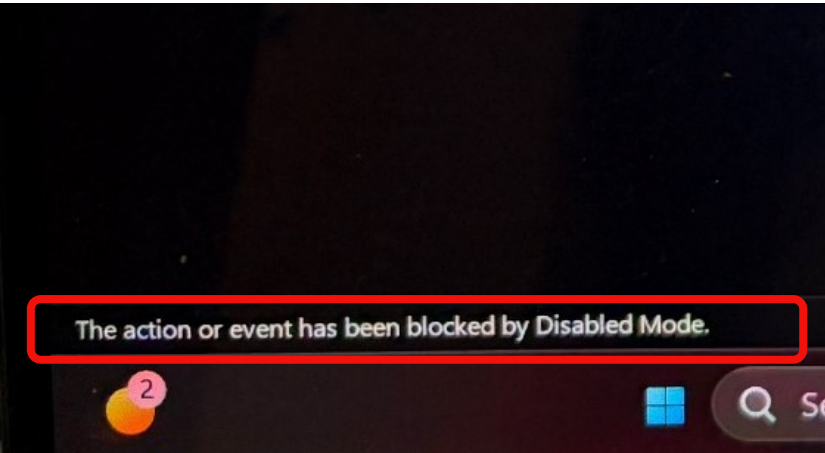
Lab PC - Microsoft Access

Use Library or Lab PCs to access Microsoft Access

In Windows >> Search for Access and Open the App

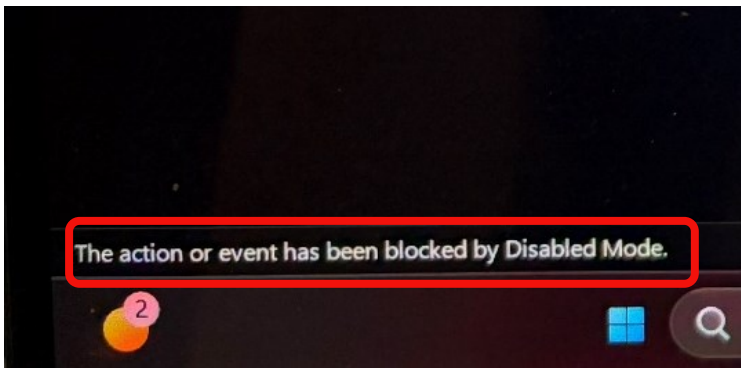


Enable Content in Access (Quick Fix Guide)



This is a very common Microsoft Access issue—nothing is actually “broken,” it’s a security setting.

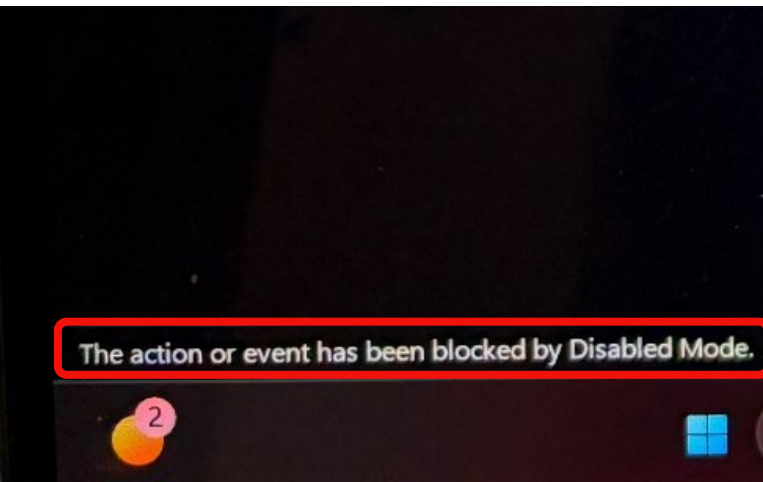
Enable Content in Access (Quick Fix Guide)



The action or event has been blocked by Disabled Mode.

Access is blocking your file because it thinks it might be unsafe.

Enable Content in Access (Quick Fix Guide)



This happens a lot in labs because:

- Files come from USB / downloads / LMS
- Different PCs = not trusted locations

Enable Content in Access (Quick Fix Guide)

What's happening

Access has opened the database in Disabled Mode because it doesn't fully trust the file (usually because it was downloaded, emailed, or opened from a different location).

That's why you see:

"The action or event has been blocked by Disabled Mode."

Enable Content in Access (Quick Fix Guide)

Look at the top of Access window (just below the ribbon).

You should see a yellow security warning bar.

Click:

“Enable Content”

That usually fixes it immediately.

Enable Content in Access (Quick Fix Guide)

If you doesn't see the yellow bar

Then try this:

Option 1 (quick fix if it is blocked)

- Close the database
- Right-click the file (.accdb)
- Click Properties
- Tick “Unblock” (if available)
- Click Apply → Open again

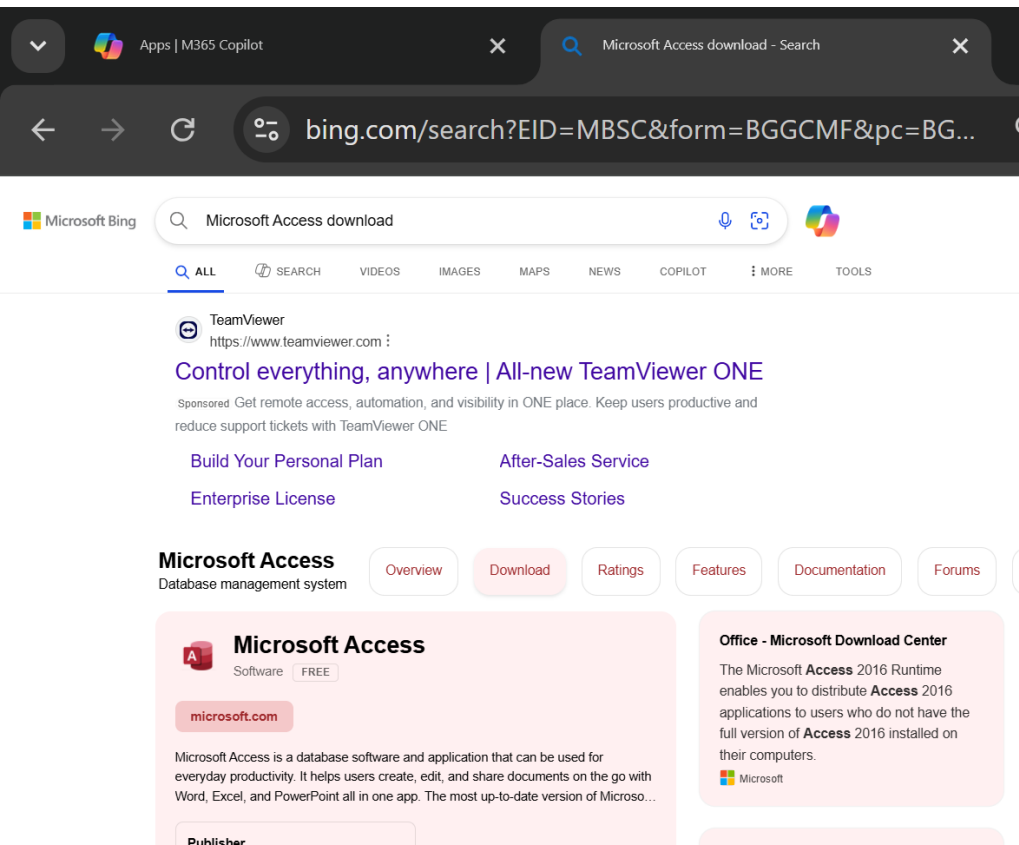
Enable Content in Access (Quick Fix Guide)

Option 2 (long-term)

- Open Access
- Go to File → Options → Trust Center
- Click Trust Center Settings
- Go to Trusted Locations
- Add the folder where you save your files (e.g., Desktop or Documents)

If You Want to Install Microsoft Access On Your Windows Laptop

1. In a private window (ctrl + shift + New)
2. Search for Microsoft Access Download
3. Click on Download

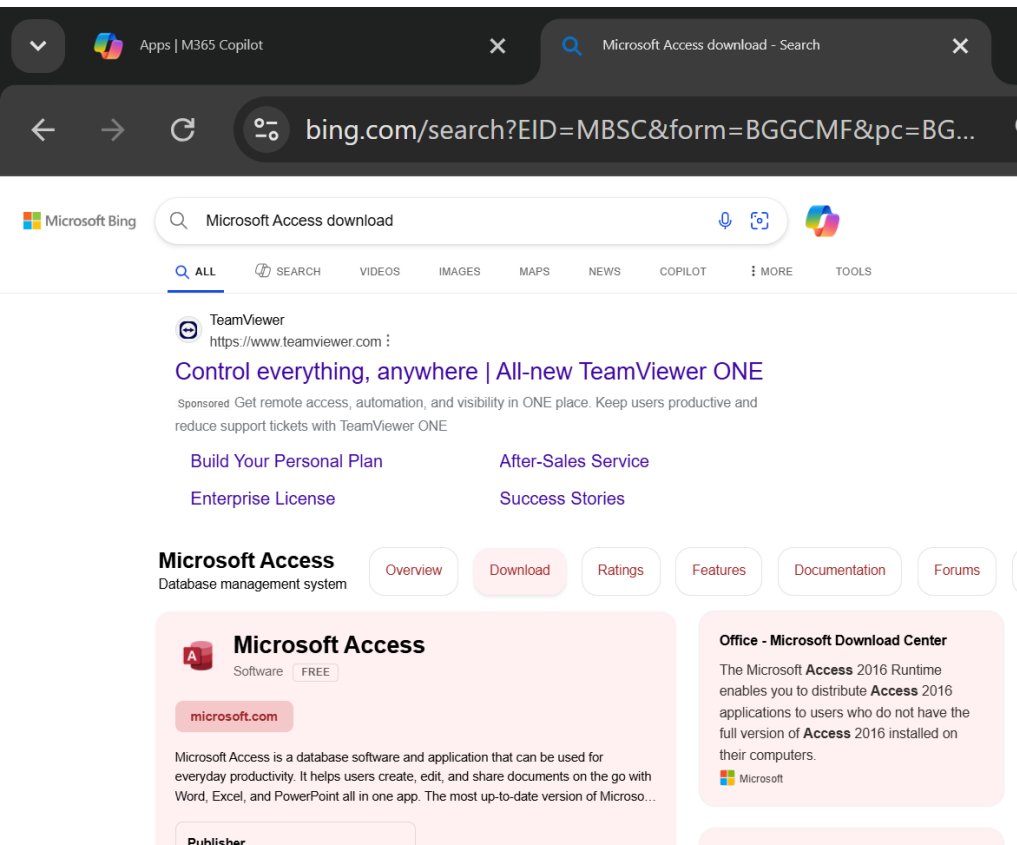


System requirements include:

- Windows 10/11
- 1.6 GHz, 2-core processor
- 4 GB RAM (64-bit)
- 4 GB disk space

If You Want to Install Microsoft Access On Your Windows Laptop

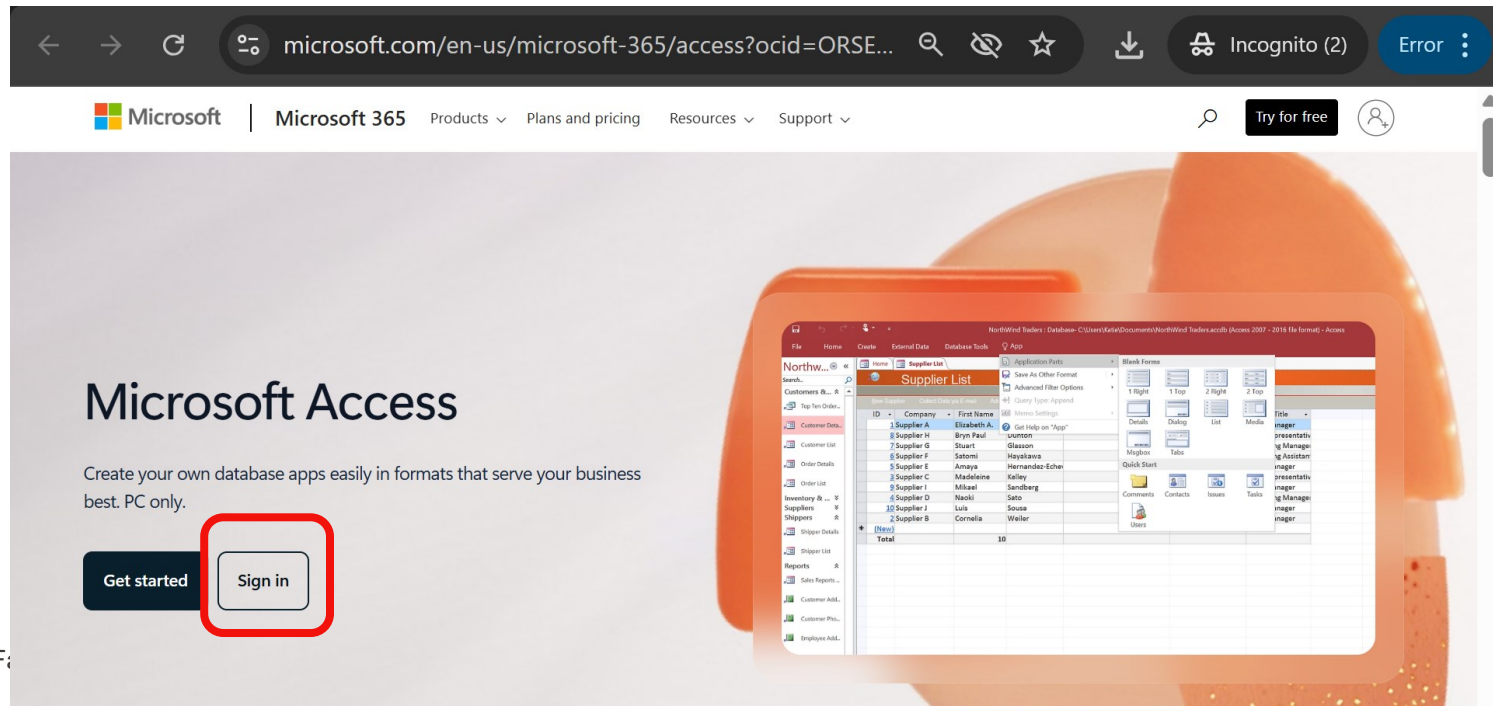
1. In a private window (ctrl + shift + New)
2. Search for Microsoft Access Download
3. Click on Download



Mac users: Microsoft Access is Windows-only. Use a library or PC computer to use Access

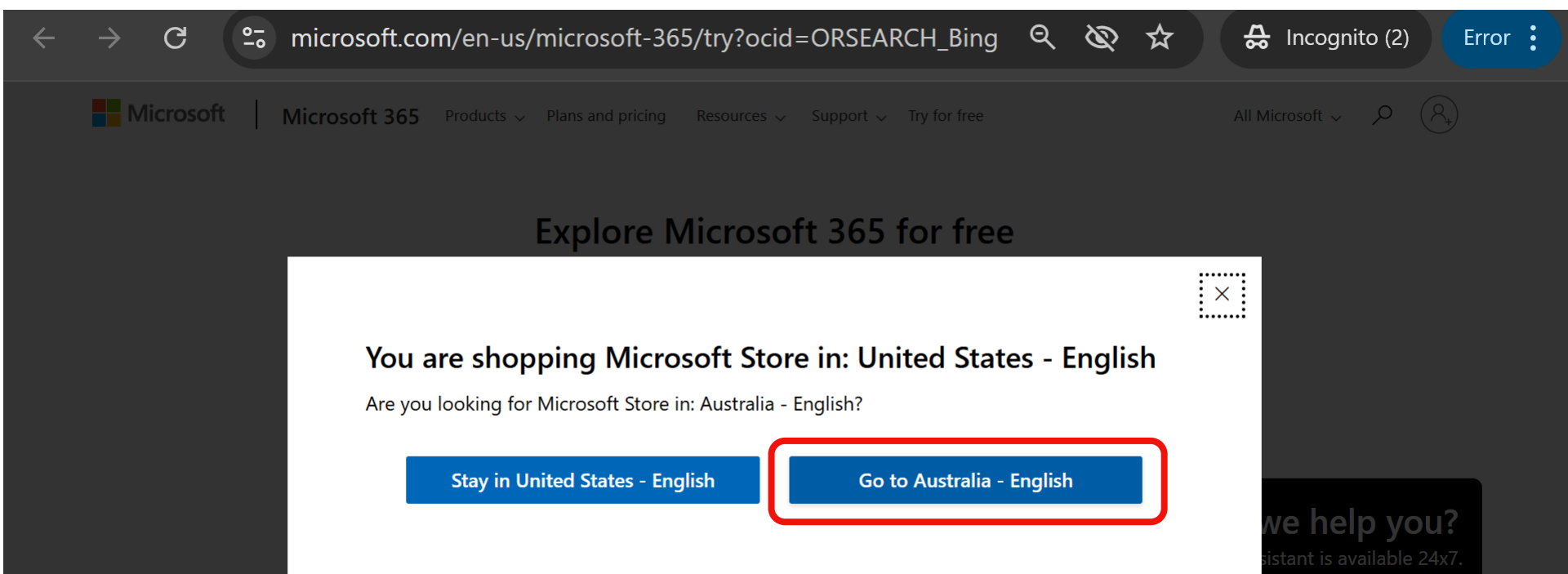
If You Want to Install Microsoft Access On Your Windows Laptop

4. Click “Sign in” (not *Try for free*). Use your ACU student email (e.g., StudentID@myacu.edu.au).
- “Try for free” will create a personal 1-month trial, if your free ACU-licensed education account is not working.



If You Want to Install Microsoft Access On Your Windows Laptop

- If sign-in is not working, click on Try for Free and Sign up



← → ↻ 🔍 microsoft.com/en-us/microsoft-365/try?ocid=ORSEARCH_Bing 🔍 🔍 ☆ Incognito (2) Error

Microsoft | Microsoft 365 Products ▾ Plans and pricing Resources ▾ Support ▾ Try for free All Microsoft 🔍

Explore Microsoft 365 for free

×

You are shopping Microsoft Store in: United States - English

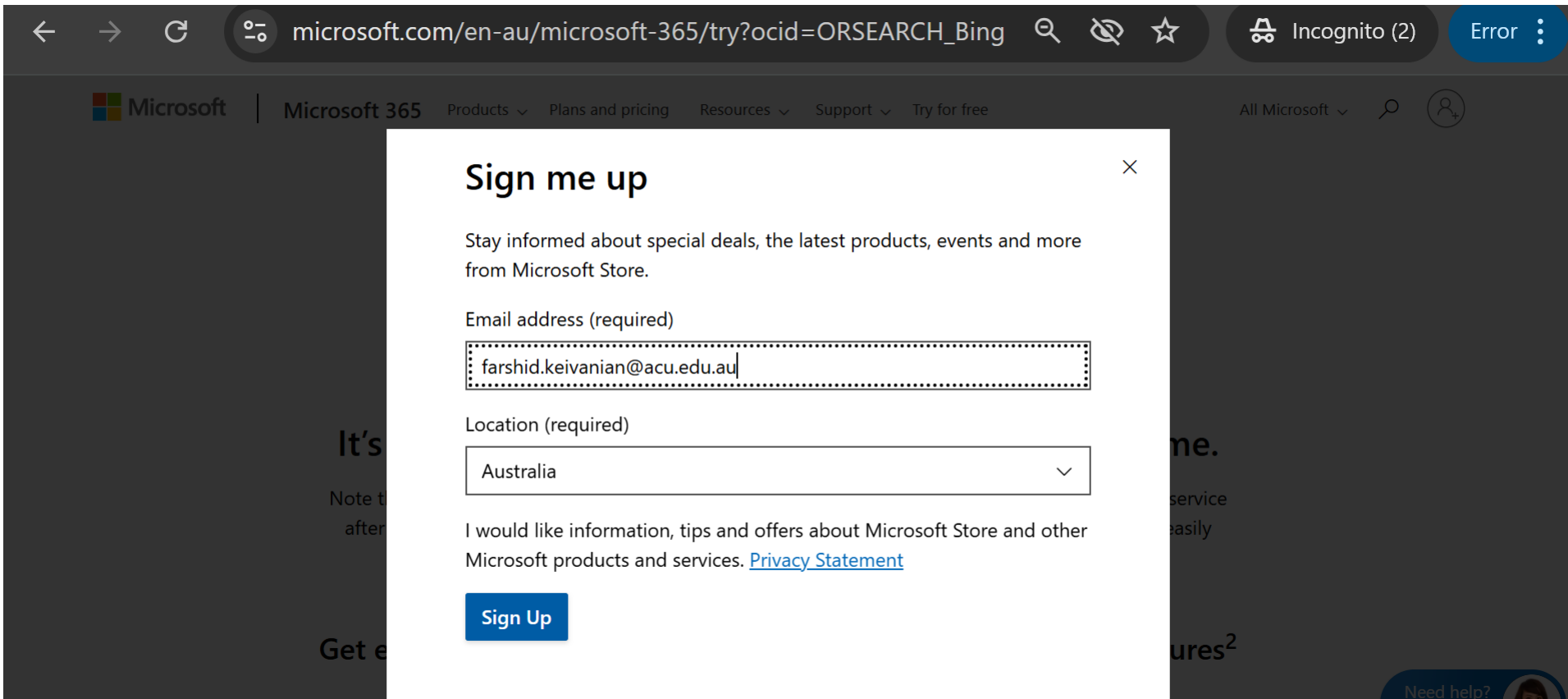
Are you looking for Microsoft Store in: Australia - English?

[Stay in United States - English](#) [Go to Australia - English](#)

we help you?
Assistant is available 24x7.

If You Want to Install Microsoft Access On Your Windows Laptop

- Enter ACU email



The screenshot shows a web browser window with the URL `microsoft.com/en-au/microsoft-365/try?ocid=ORSEARCH_Bing`. The browser is in Incognito mode. The Microsoft 365 website header is visible, including the Microsoft logo and navigation links. A 'Sign me up' modal is displayed in the center of the screen. The modal contains the following text and form elements:

Sign me up [Close icon]

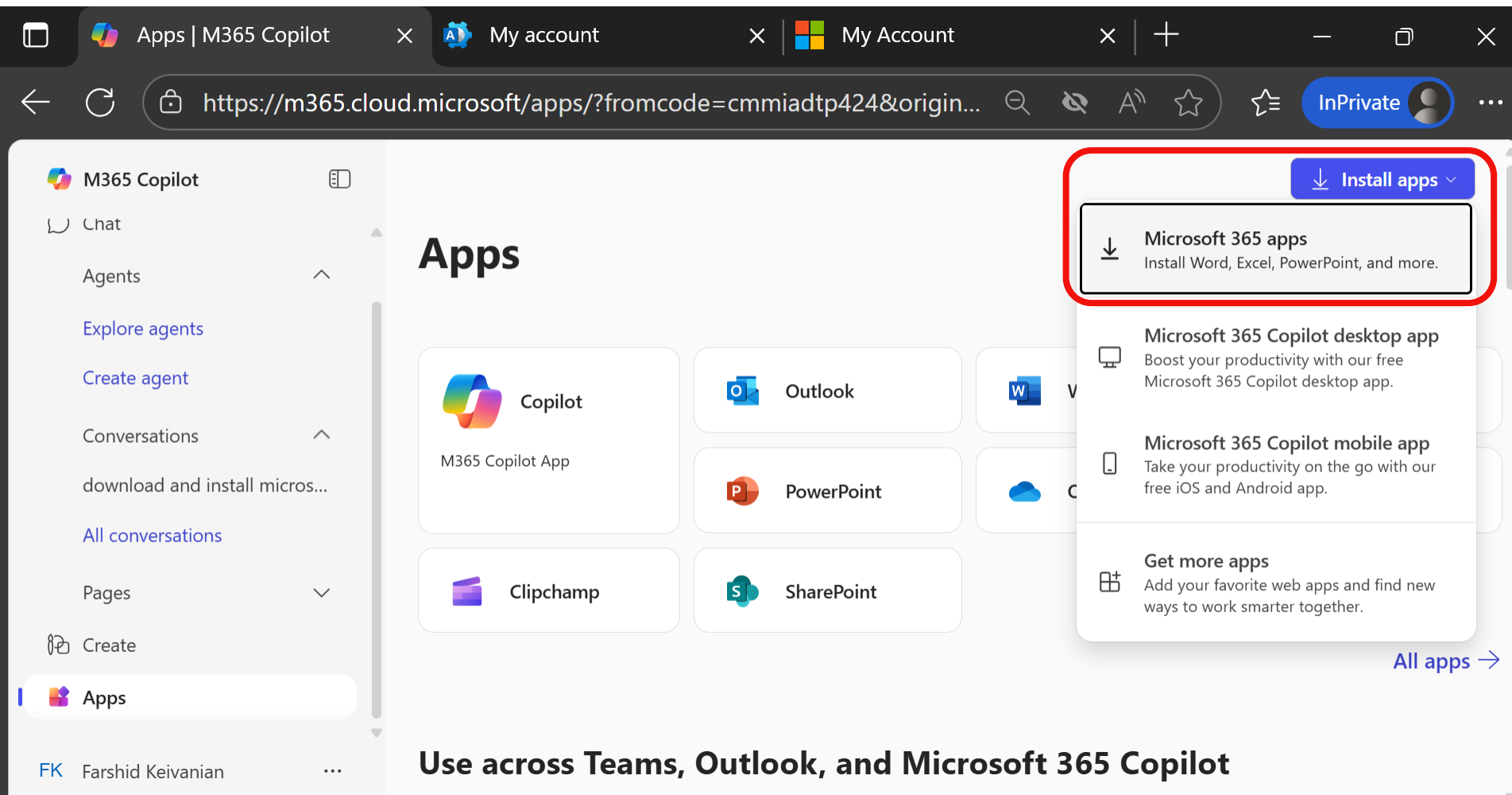
Stay informed about special deals, the latest products, events and more from Microsoft Store.

Email address (required)

Location (required)

I would like information, tips and offers about Microsoft Store and other Microsoft products and services. [Privacy Statement](#)

If You Want to Install Microsoft Access On Your Windows Laptop



Apps | M365 Copilot × My account × My Account × + - □ ×

https://m365.cloud.microsoft/apps/?fromcode=cmmiadtp424&origin... 🔍 🔄 ⬅️ ☆ InPrivate

M365 Copilot

- Chat
- Agents
- Explore agents
- Create agent
- Conversations
- download and install micros...
- All conversations
- Pages
- Create
- Apps**

Apps

-  **Copilot**
M365 Copilot App
-  **Outlook**
-  **PowerPoint**
-  **Clipchamp**
-  **SharePoint**

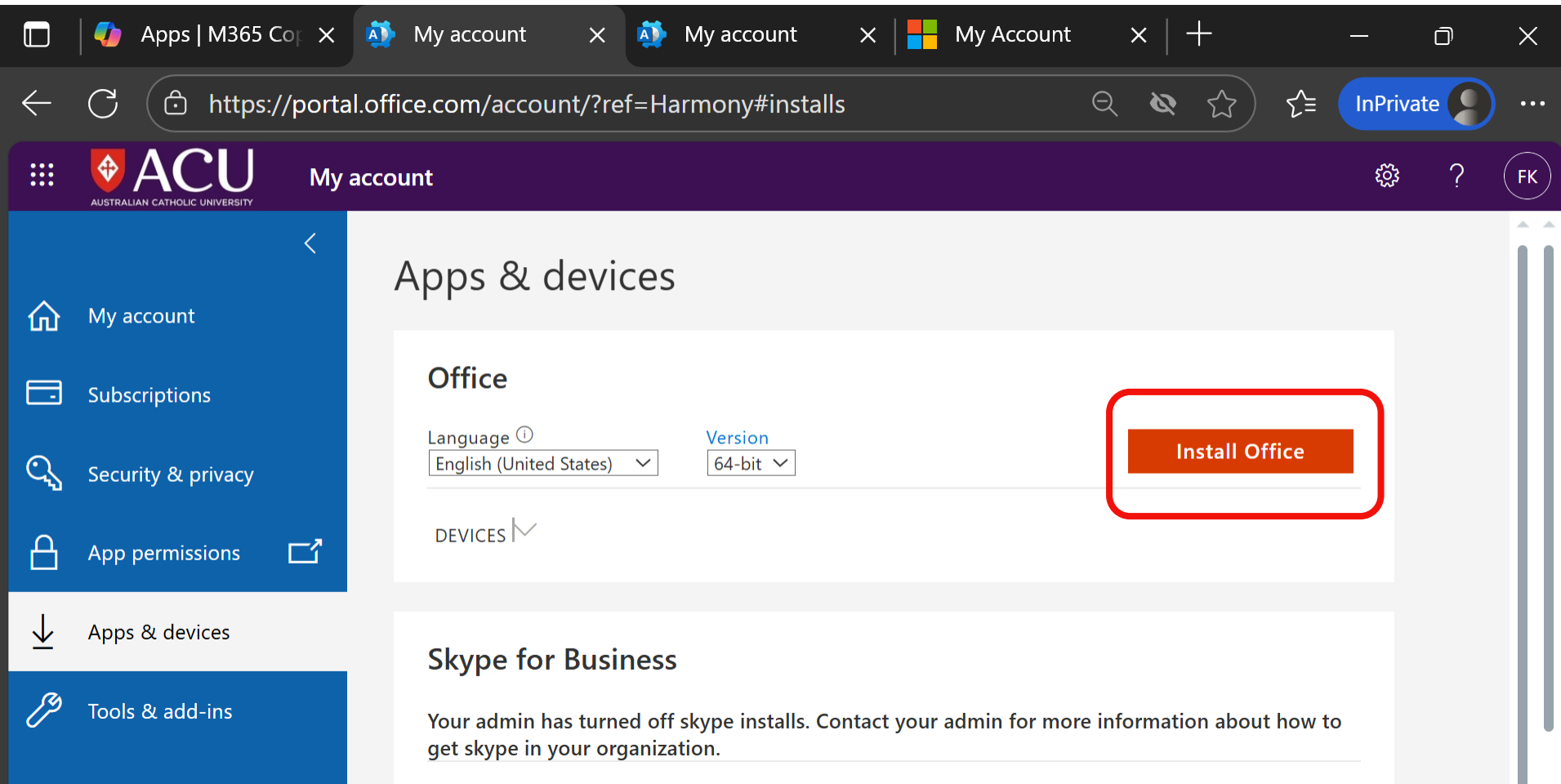
Install apps

-  **Microsoft 365 apps**
Install Word, Excel, PowerPoint, and more.
-  **Microsoft 365 Copilot desktop app**
Boost your productivity with our free Microsoft 365 Copilot desktop app.
-  **Microsoft 365 Copilot mobile app**
Take your productivity on the go with our free iOS and Android app.
-  **Get more apps**
Add your favorite web apps and find new ways to work smarter together.

[All apps →](#)

Use across Teams, Outlook, and Microsoft 365 Copilot

If You Want to Install Microsoft Access On Your Windows Laptop



The screenshot shows a web browser window with multiple tabs. The active tab is titled "My account" and the address bar shows the URL "https://portal.office.com/account/?ref=Harmony#installs". The page header is purple and contains the ACU logo, the text "My account", and a user profile icon labeled "FK". The left sidebar is blue and contains links to "My account", "Subscriptions", "Security & privacy", "App permissions", "Apps & devices" (which is selected), and "Tools & add-ins". The main content area is titled "Apps & devices" and contains two sections. The "Office" section has a red box around the "Install Office" button. Below it, the "DEVICES" section is partially visible. The "Skype for Business" section has a message: "Your admin has turned off skype installs. Contact your admin for more information about how to get skype in your organization."

Apps | M365 Co... x My account x My account x My Account x +

https://portal.office.com/account/?ref=Harmony#installs

ACU AUSTRALIAN CATHOLIC UNIVERSITY My account

Apps & devices

Office

Language ⓘ English (United States) Version 64-bit

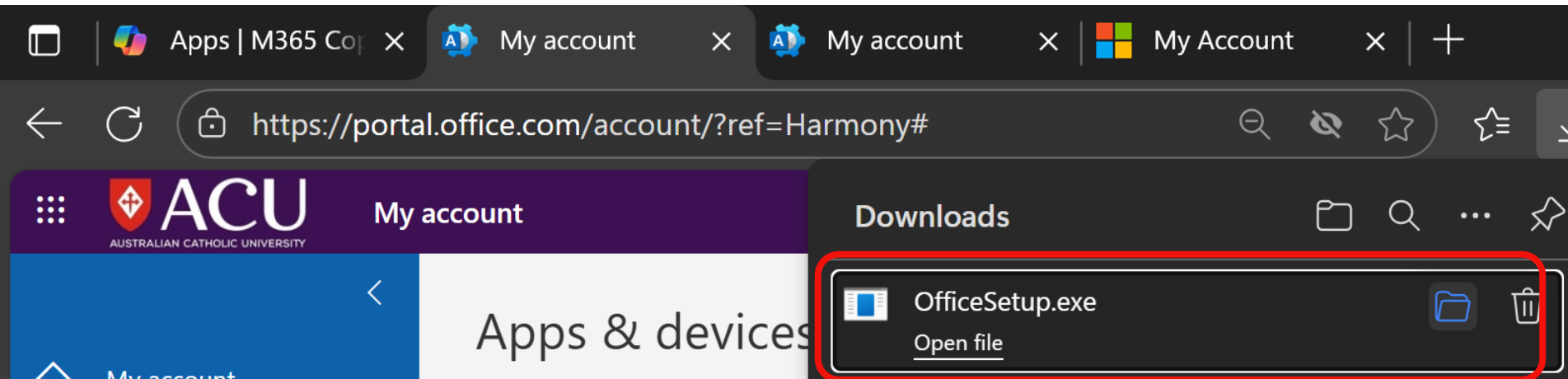
Install Office

DEVICES

Skype for Business

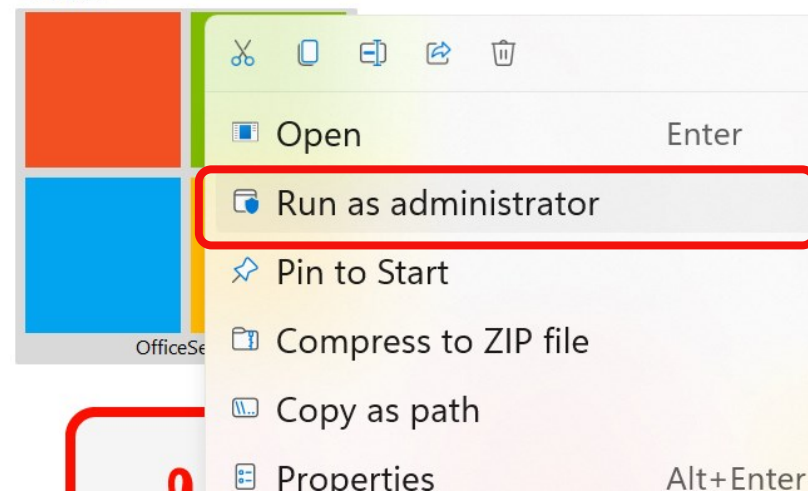
Your admin has turned off skype installs. Contact your admin for more information about how to get skype in your organization.

If You Want to Install Microsoft Access On Your Windows Laptop



sk (C:) > Users > gaming 15 > Downloads

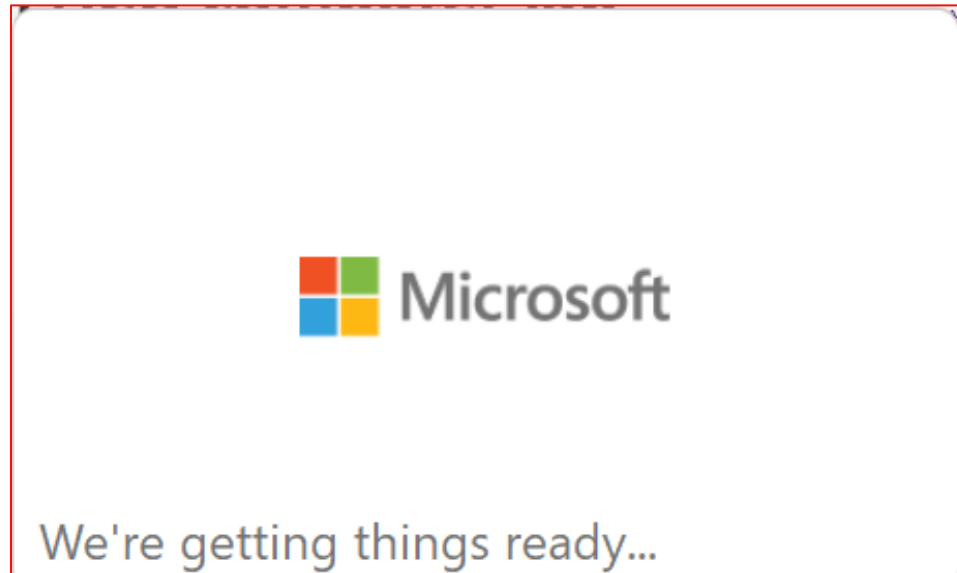
Today (8)



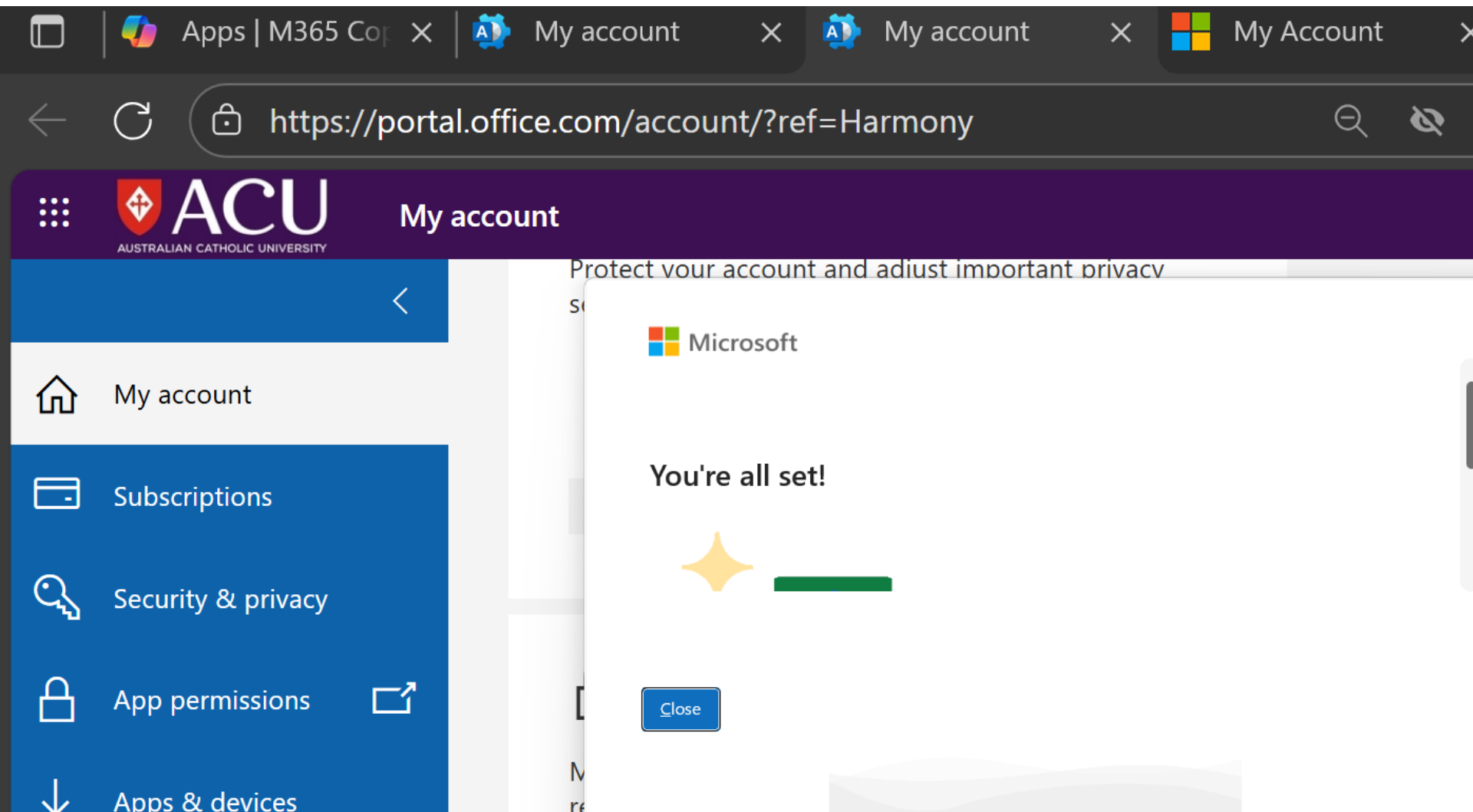
Right Click and Run
as administrator

School

If You Want to Install Microsoft Access On Your Windows Laptop



If You Want to Install Microsoft Access On Your Windows Laptop



Apps | M365 Co | X | My account X | My Account X

https://portal.office.com/account/?ref=Harmony

ACU AUSTRALIAN CATHOLIC UNIVERSITY My account

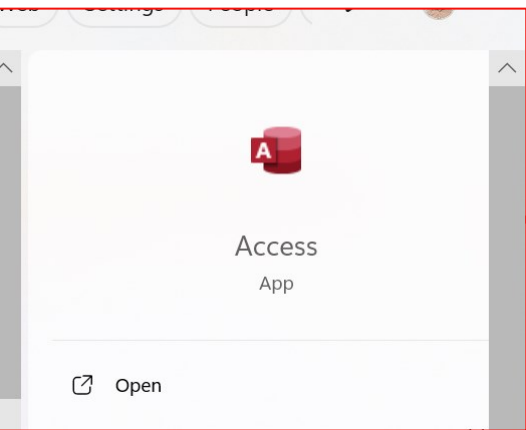
Protect your account and adjust important privacy

Microsoft

You're all set!

Close

Run Microsoft Access



Access

Home

New

Open

Good afternoon

New

Blank database

Northwind starter edition

Northwind dev edition

Asset tracking

Contacts

Students

More templates →

Search

Recent

Favorites

You haven't opened any files recently. Click Open to browse for a file.

Farshid Keivanian

?

—

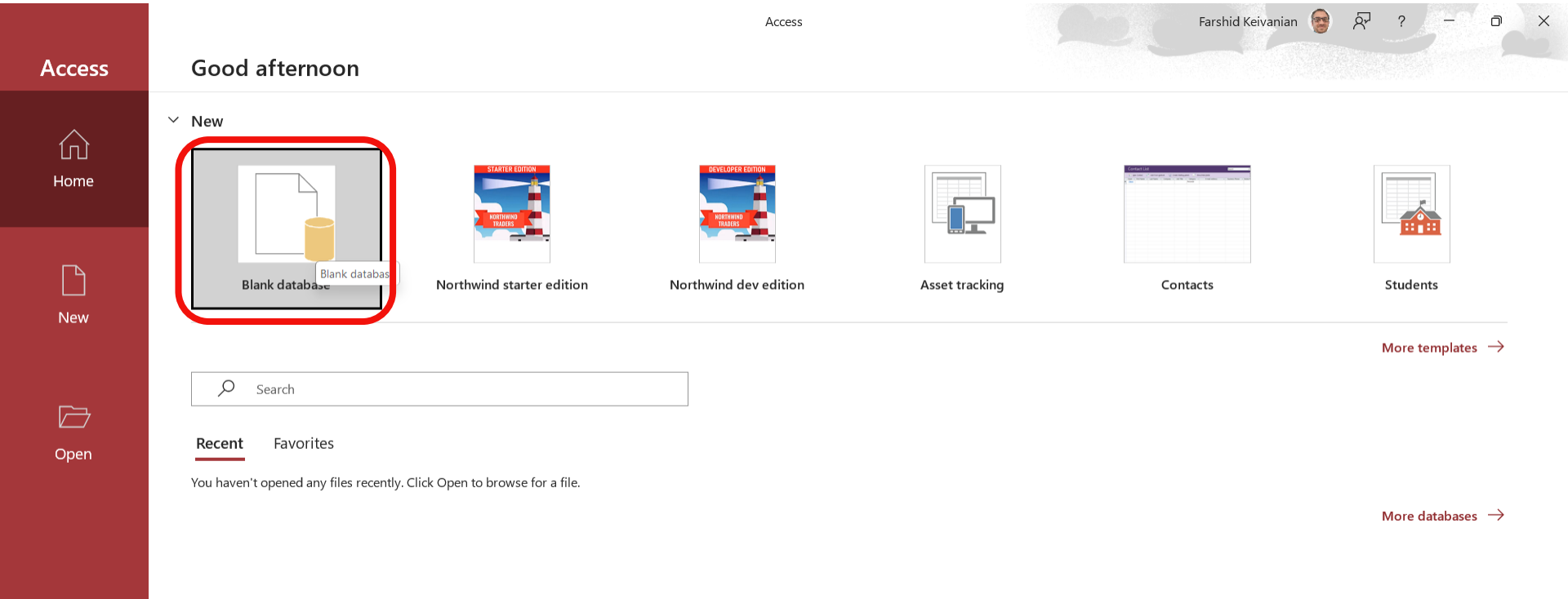
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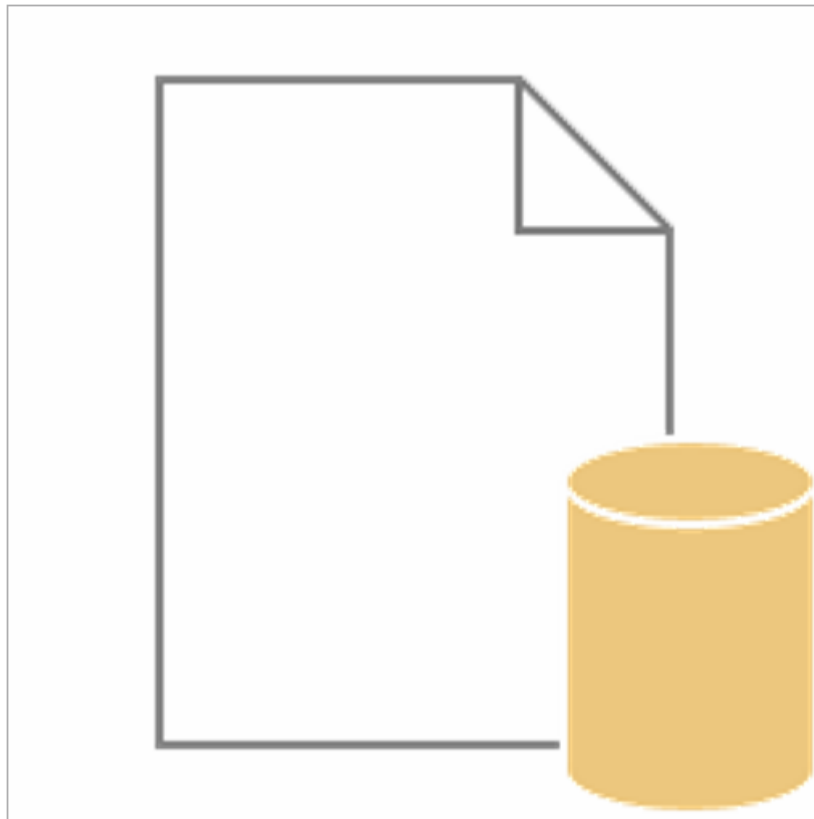
More databases →

30 | Faculty of Business and Law | Peter Faber Business School

Run Microsoft Access



Run Microsoft Access



Blank database

File Name

Week5

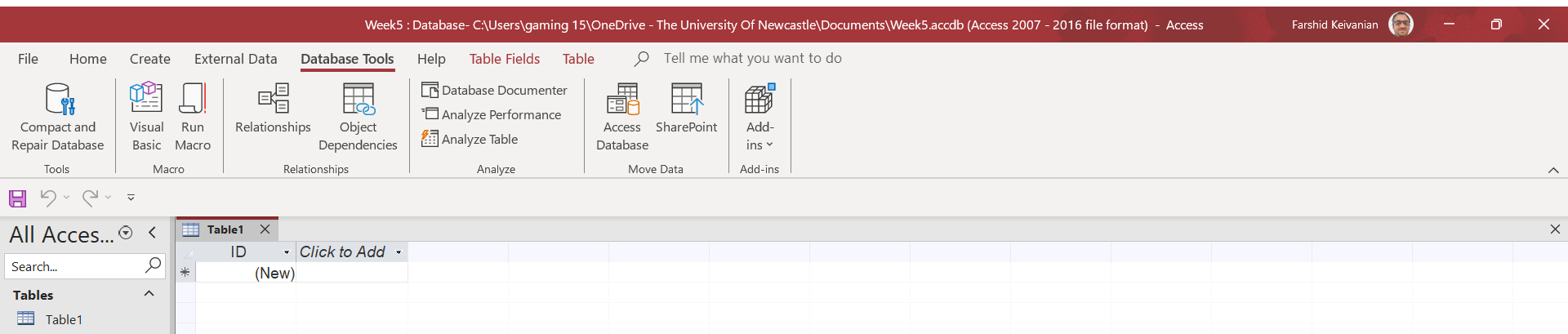


C:\Users\gaming 15\OneDrive - The University Of
Newcastle\Documents\



Create

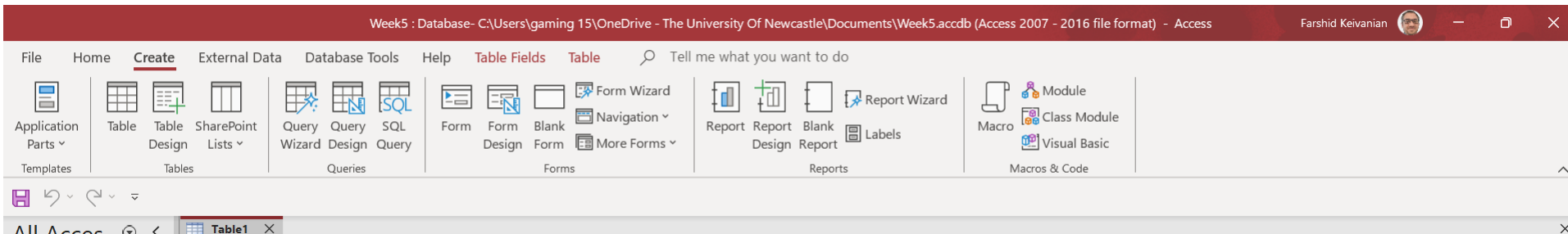
Menus in Microsoft Access



Database Tools Tab

- Tools to manage your whole database.
- Relationships → connect tables together (e.g., Students ↔ Enrolments).
- Analyze Performance → check if your database is efficient.
- Compact and Repair Database → fix issues if the file gets too large or corrupted

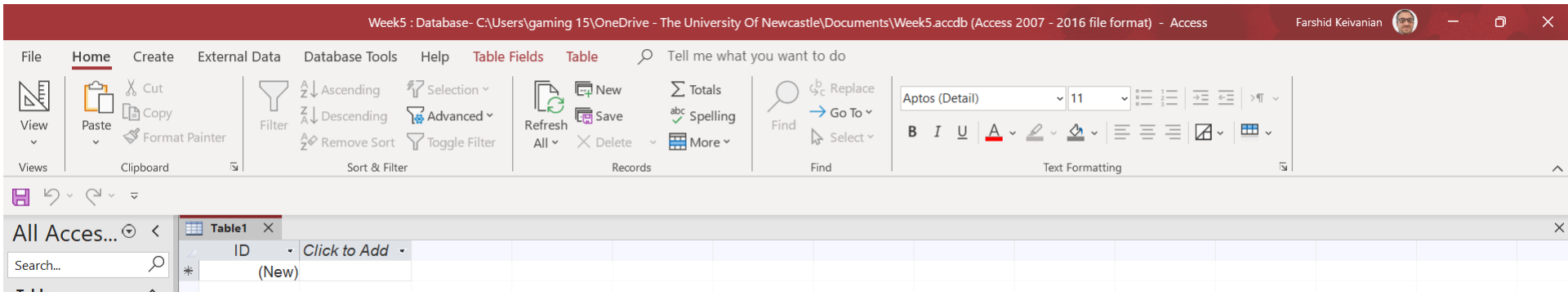
Menus in Microsoft Access



Create Tab

- Where you add new objects:
 - Table / Table Design → create new tables.
 - Query Wizard / Query Design → build queries to ask questions.
 - Form / Form Wizard → make forms for easier data entry.
 - Report / Report Wizard → create reports to print or share summaries.

Menus in Microsoft Access



Home Tab

- Everyday editing tools.
- Clipboard → copy, paste, cut.
- Sort & Filter → arrange or filter table records.
- Records group → add new record, delete, save.
- Find → search inside the table.
- Formatting options for text (like bold, size, font).

Menus in Microsoft Access

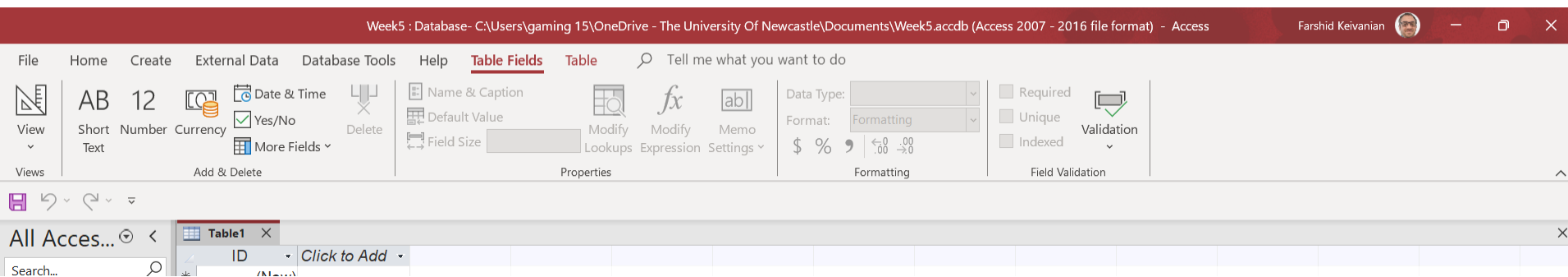


Table Fields Tab

- Settings for each field (column).
- Add new field types quickly (Short Text, Number, Currency, Yes/No, Date/Time).
- Properties → Default value, Required, Indexed.
- Validation → enforce rules (e.g., PostCode must be 4 digits).

Menus in Microsoft Access

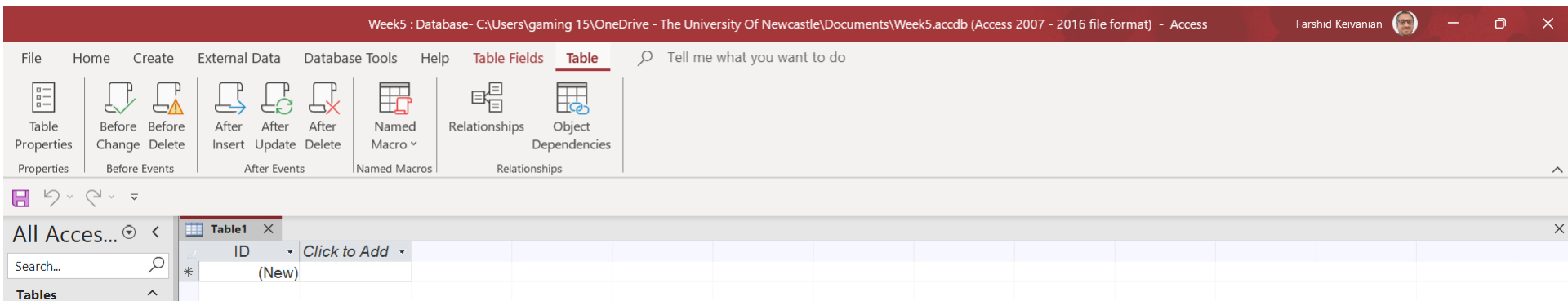
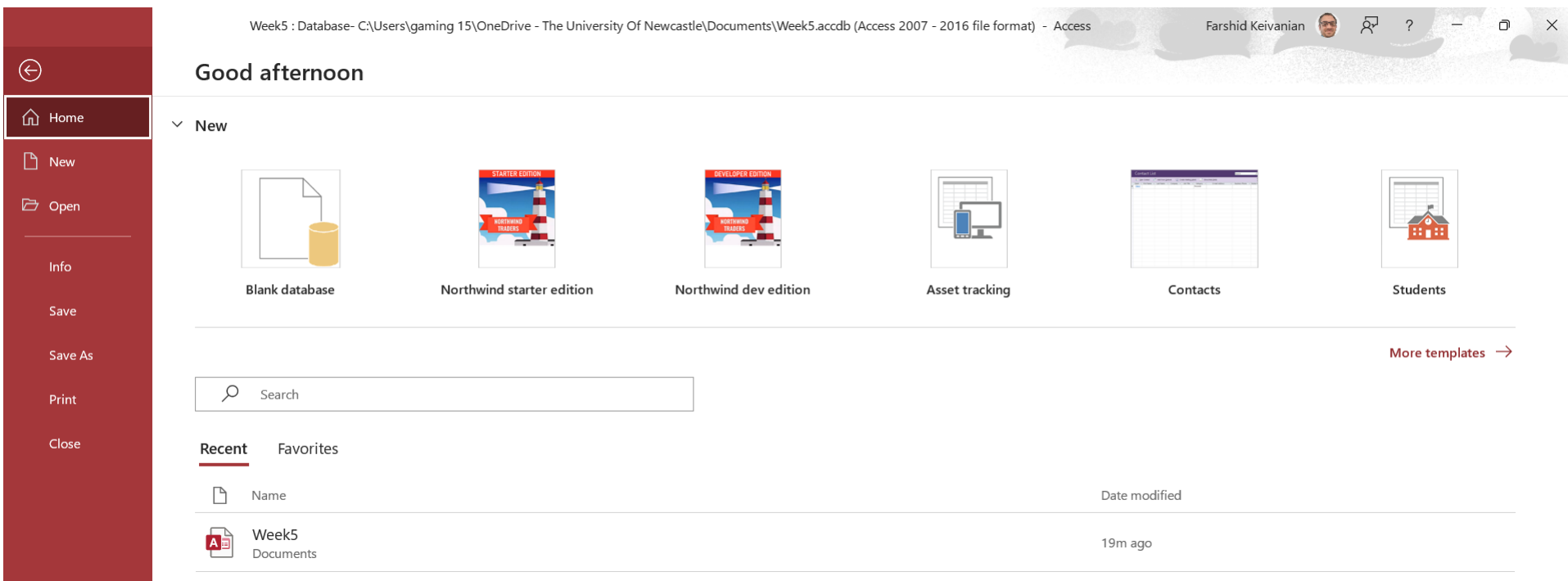


Table Tab

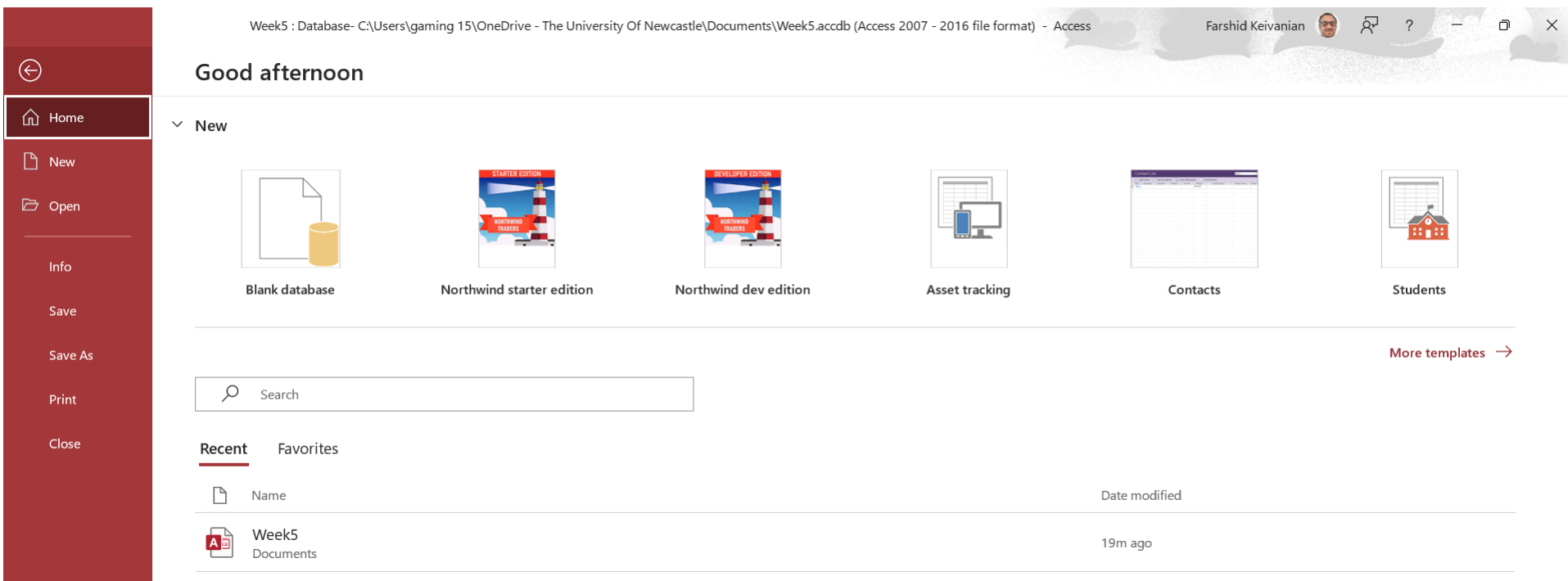
- Options that apply to the whole table.
- View → switch between Datasheet View and Design View.
- Properties → adjust settings for the table itself.
- Useful when you want to manage structure rather than just the fields.

Menus in Microsoft Access



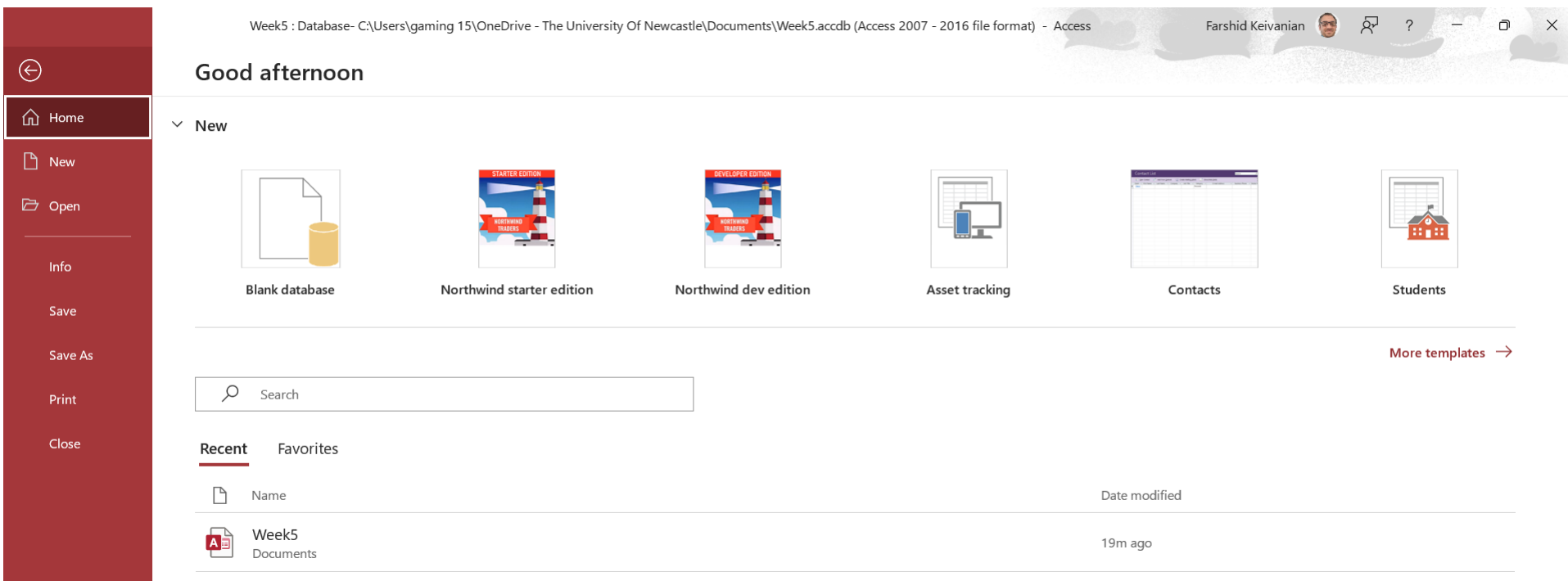
- **Home** → Shows templates (Blank database, Northwind sample DB, etc.) and your recently opened databases.
- **New** → Start a new database (blank or from a template).
- **Open** → Open existing databases from your computer, OneDrive, or network.

Menus in Microsoft Access



- **Info** → See details about the current database (size, path, permissions).
- **Save / Save As** → Save your current work or export a copy in a different format (older Access version, ACCDB to MDB, etc.).
- **Print** → Print objects (like a report or query result).

Menus in Microsoft Access



- **Close** → Close the current database file.
- **Account** → Manage your Microsoft account login (switch between ACU email or personal account).
- **Options** → Access advanced settings (default font, database options, proofing, startup behavior, etc.).

Menus in Microsoft Access



Summary:

- Home = daily editing.
- Create = build new things (tables, queries, forms, reports).
- Database Tools = manage & connect your database.
- Table Fields = manage details of each column.
- Table = settings for the whole table.
- File = Think of the File tab like the settings and file cabinet of Access — it doesn't edit your tables directly, but it lets you open, save, print, or change how Access itself works.

- **Tables** = store data.
- **Queries** = ask questions about the data.
- **Forms** = user-friendly data entry.
- **Reports** = printable summaries.
- **Macros & Modules** = automation.

Example (Australia):

- **Table:** e.g., Student data at ACU.
- **Query:** e.g., List all students from NSW.
- **Report:** e.g., Generate student attendance report.

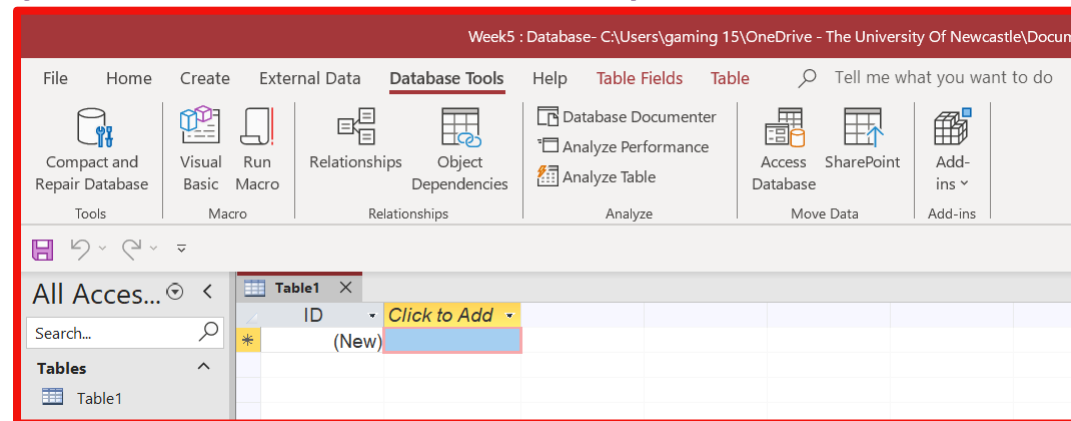
MS Access Environment (Ribbons & Navigation Pane)

- Ribbon tabs = tools grouped.
- Navigation Pane = like “Contents Page” of your DB.
- Trust Center = approve your file to run safely.

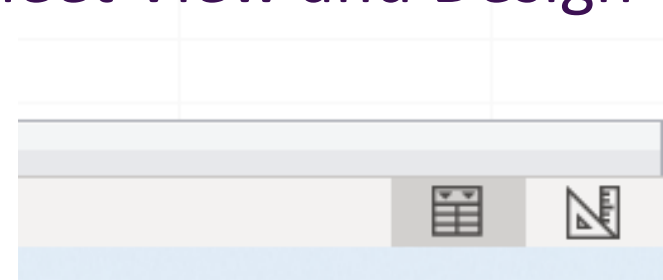
Hands-on Activity:

- Explore how to collapse/expand

Navigation Pane with **F11**.



Identify where to switch between Datasheet View and Design View.



Two views:

- **Datasheet View** → like Excel, quick entry.
- **Design View** → control field names, data types, keys.

Analogy:

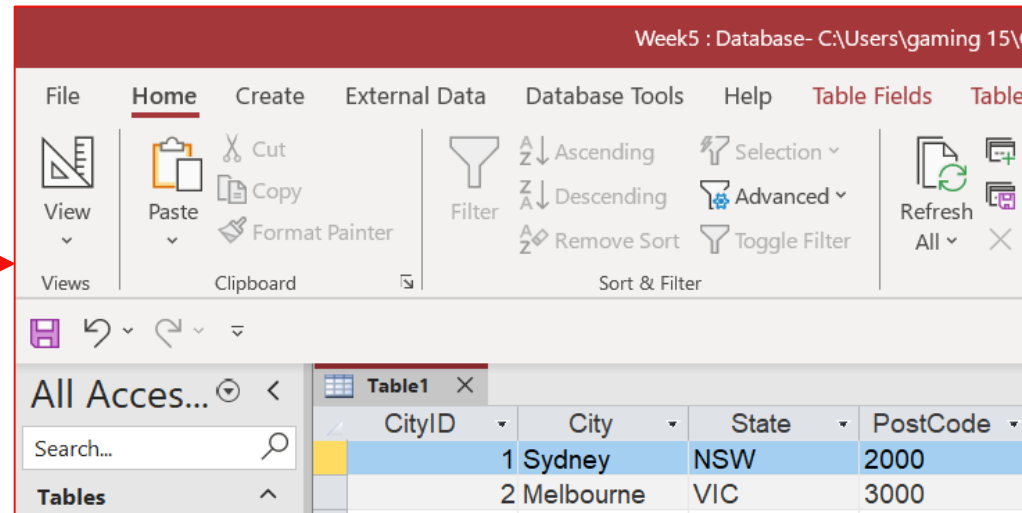
Datasheet = entering data into a form.

Design = creating the blank form with rules.

Example:

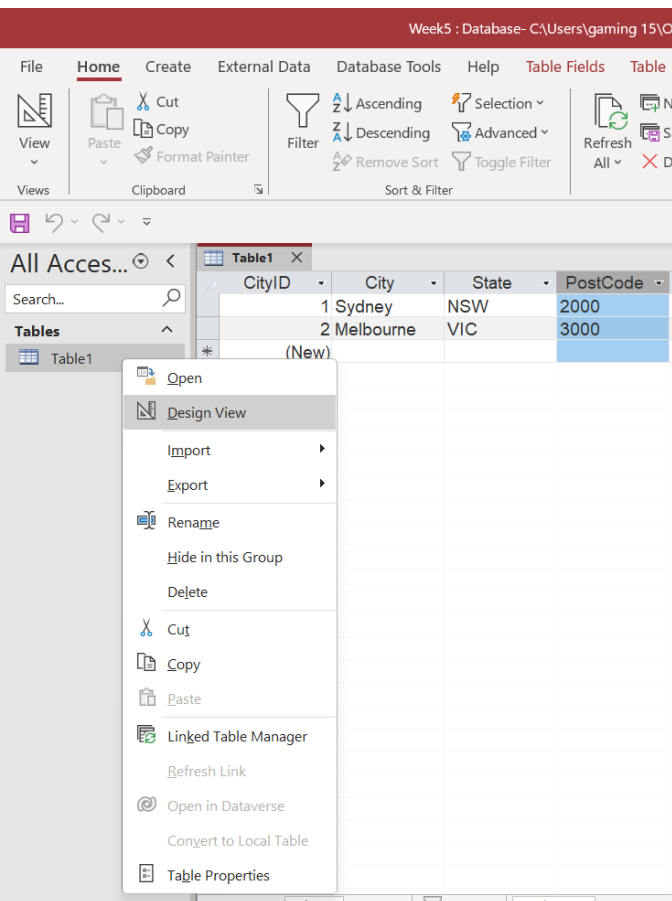
Create a City table for Australian cities.

CityID	City	State	PostCode
1	Sydney	NSW	2000
2	Melbourne	VIC	3000

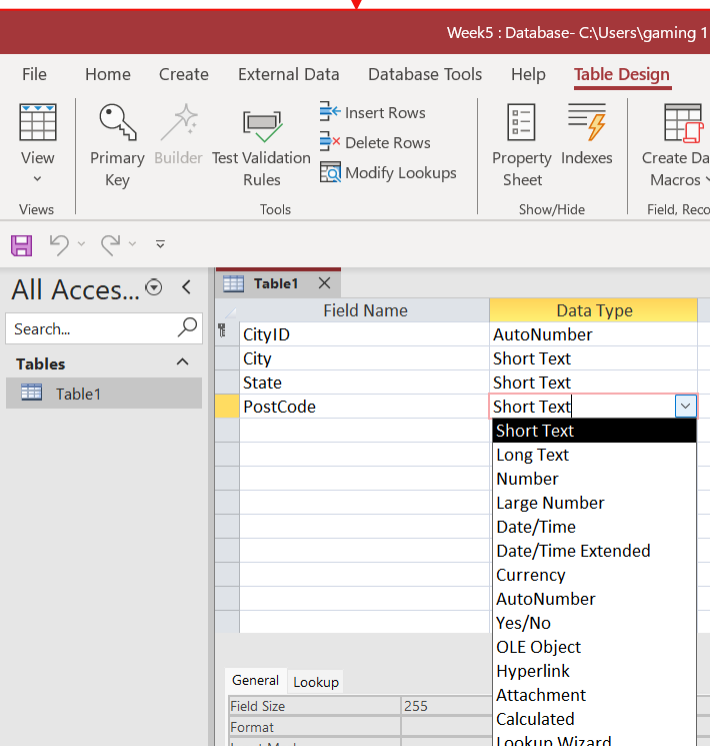
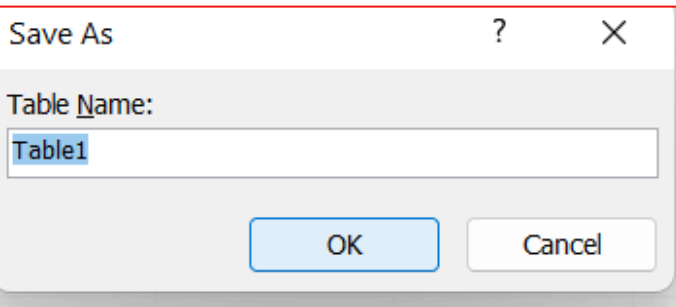


Hands-on Activity:

- Ensure PostCode is text, not number.
- Right-click on *Table1* (from the left panel under "Tables").
- Select **Design View**.



Data Types in Access

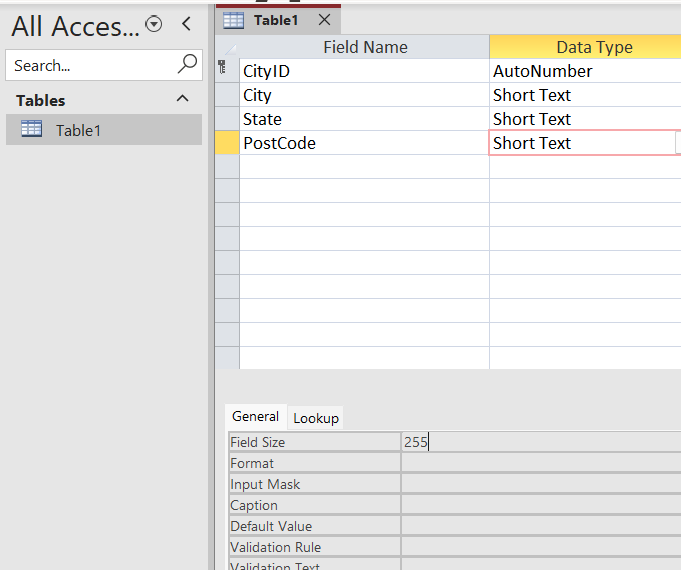


- Select Table Name
- Locate the PostCode field in the list of fields.
- In the Data Type column (next to PostCode), change it from:
Number → to Short Text.



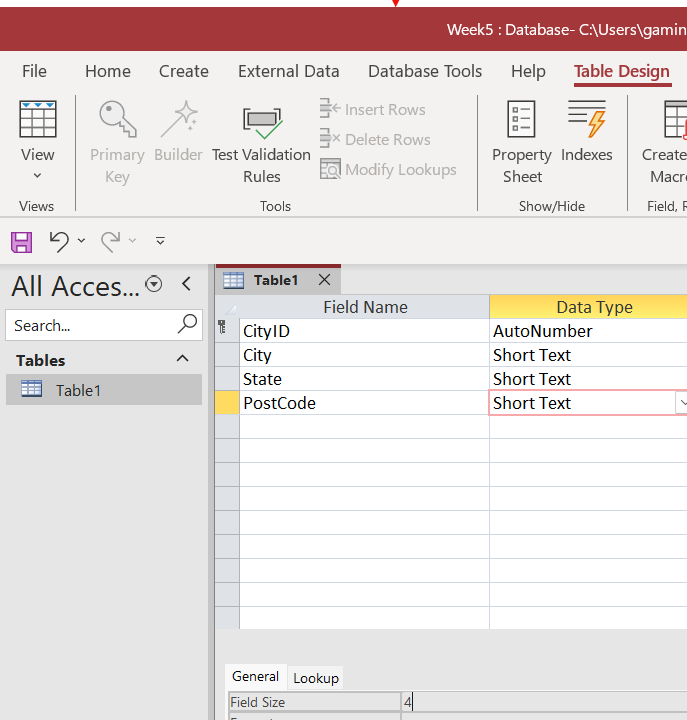
Field Name	Data Type
CityID	AutoNumber
City	Short Text
State	Short Text
PostCode	Short Text

Data Types in Access



Set a Field Size:

- In the bottom "Field Properties" section, set Field Size = 4 (since Australian postcodes are 4 digits).
- This prevents users from entering too many characters.
- Save the table (Ctrl + S).



Activity 1: Building a Table and Adding Rules to Fields

Goal: Students will create a table, add a few rows of data, and then apply rules to make the data more reliable and meaningful.

Activity 1: Building a Table and Adding Rules to Fields

Step 1: Create a Table

1. Open MS Access → **Create** tab → **Table Design**.
2. Add these fields:
 - **PatientID** → *AutoNumber* (Primary Key)
 - **FirstName** → *Short Text*
 - **LastName** → *Short Text*
 - **State** → *Short Text*
 - **PostCode** → *Short Text*
 - **Balance** → *Currency*

Save the table as **Patient**.

Activity 1: Building a Table and Adding Rules to Fields

Step 2: Enter Sample Data (2 Rows)

Switch to **Datasheet View** and type:

PatientID	FirstName	LastName	State	PostCode	Balance
1	Farshid	Keivanian	NSW	2000	5000
2	Mary	Brown	VIC	3000	12000

Week5 : Database- C:\Users\gaming 15\OneDrive - The University Of Newcastle\Documents\W

File Home Create External Data Database Tools Help Table Fields Table Tell me what you want to do

View Paste Cut Copy Format Painter Filter Sort & Filter Filter Ascending Descending Remove Sort Advanced Toggle Filter Refresh New Save Delete More Find Replace Go To Select

Tables

PatientID	FirstName	LastName	State	PostCode	Balance	Click to Add
1	Farshid	Keivanian	NSW	2000	\$5,000.00	
2	Mary	Brown	VIC	3000	\$12,000.00	

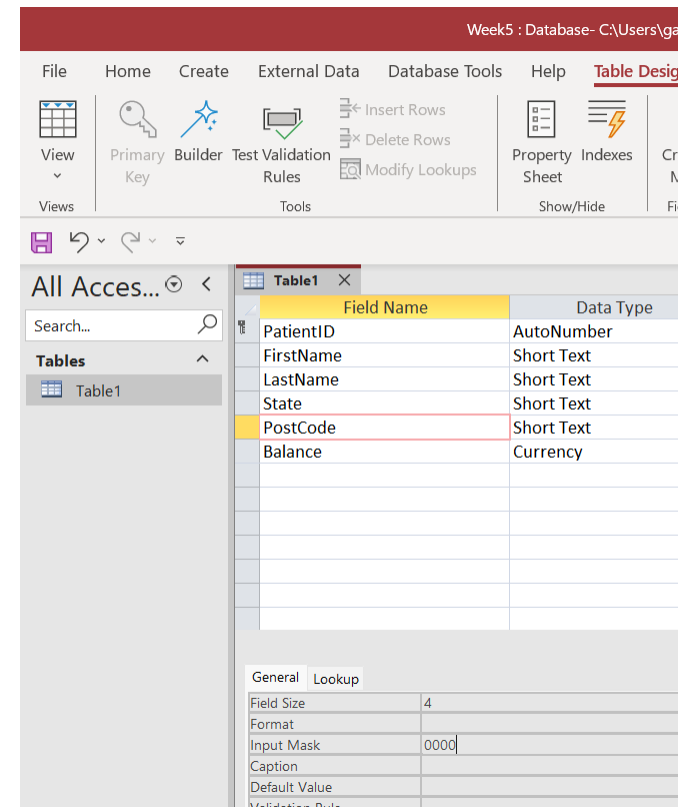
Activity 1: Building a Table and Adding Rules to Fields

Step 3: Add Rules to Fields

(a) Input Mask for PostCode → 4 digits only

- Go to Design View → Select *PostCode*.
- In *Field Properties* → *Input Mask*, choose 0000.
- Test it: Students cannot enter letters or more than 4 digits.

Analogy: Like forcing a phone keypad to accept only numbers.

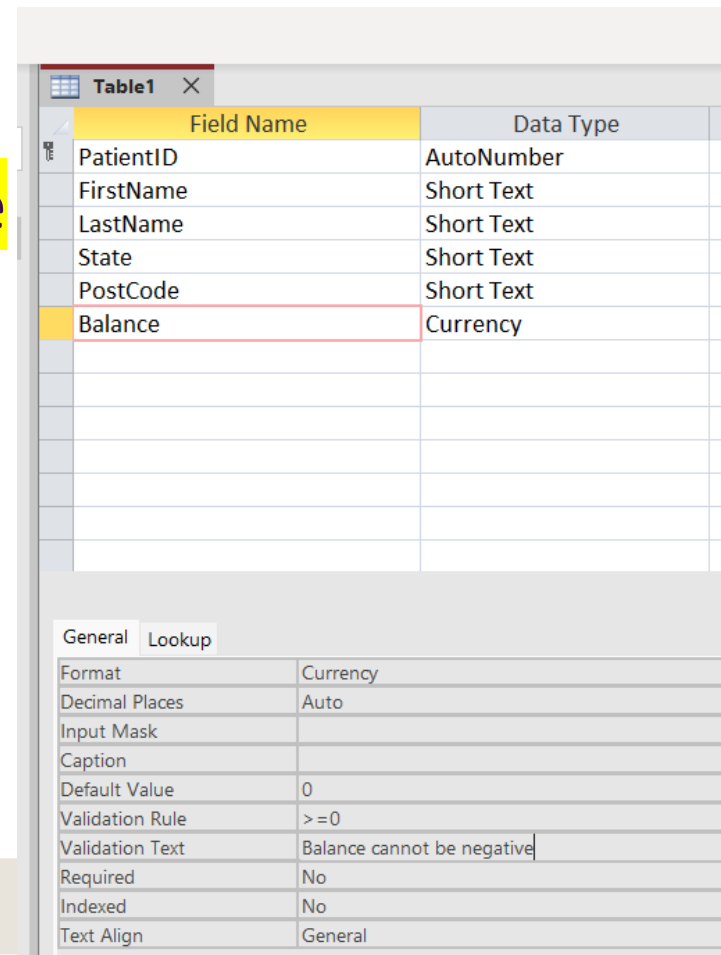


Activity 1: Building a Table and Adding Rules to Fields

(b) Validation Rule for Balance → must be ≥ 0

- Select *Balance*.
- In *Validation Rule*, type: ≥ 0
- In *Validation Text*, type: "Balance cannot be negative".

Analogy: A bank account cannot go below zero.



The screenshot shows the Microsoft Access interface. At the top, a window titled 'Table1' displays a table design view with two columns: 'Field Name' and 'Data Type'. The table contains the following fields:

Field Name	Data Type
PatientID	AutoNumber
FirstName	Short Text
LastName	Short Text
State	Short Text
PostCode	Short Text
Balance	Currency

The 'Balance' field is selected, and the Properties window is open, showing the 'General' tab. The 'Validation Rule' is set to ≥ 0 and the 'Validation Text' is 'Balance cannot be negative'.

Property	Value
Format	Currency
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	≥ 0
Validation Text	Balance cannot be negative
Required	No
Indexed	No
Text Align	General

Activity 1: Building a Table and Adding Rules to Fields

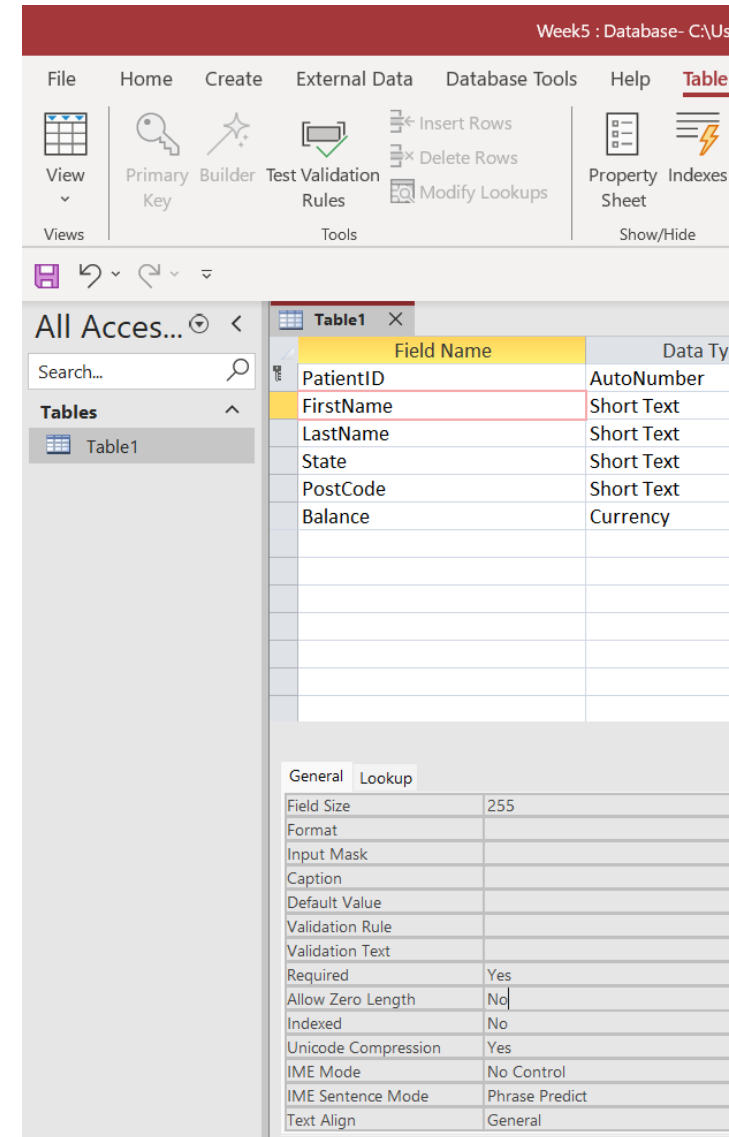
(c) Required → FirstName cannot be blank

Select *FirstName*.

In *Field Properties*, set **Required** = **Yes**.

Allow zero length = **No**.

Analogy: Every patient must have a first name; blank is not acceptable.



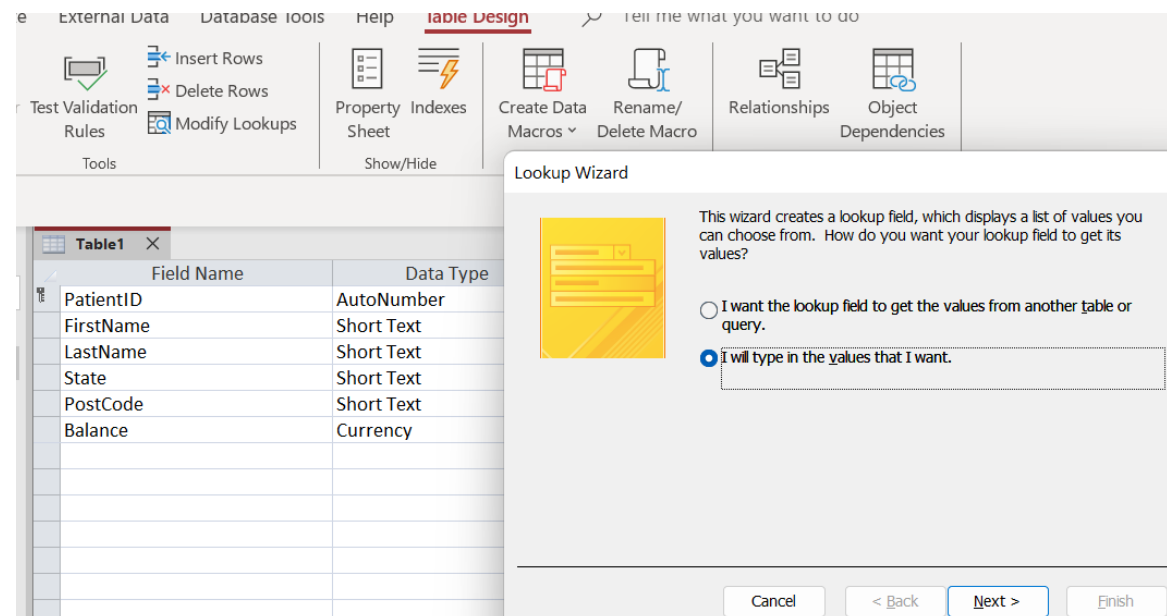
Activity 1: Building a Table and Adding Rules to Fields

(d) Lookup Wizard → Dropdown for State

Select *State*.

Data Type → **Lookup Wizard**.

Choose “I will type in the values.”



Activity 1: Building a Table and Adding Rules to Fields

Enter: NSW, VIC, QLD, SA,
WA, TAS, NT, ACT.

Analogy: Like a drop-down
list on a website that
avoids spelling mistakes.

Lookup Wizard

What values do you want to see in your lookup field? Enter the number of columns you want in the list, and then type the values you want in each cell.

To adjust the width of a column, drag its right edge to the width you want, or double-click the right edge of the column heading to get the best fit.

Number of columns:

Col1
NSW
VIC
QLD
SA
WA
TAS
NT
ACT

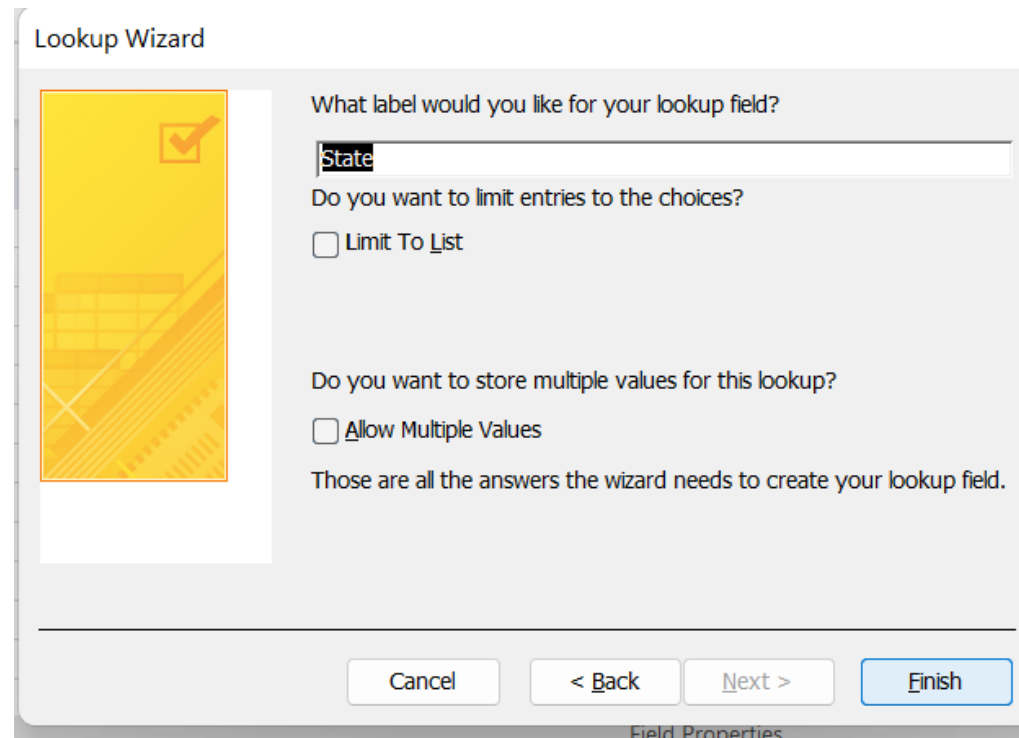
Cancel < Back Next > Finish

Field Properties

Activity 1: Building a Table and Adding Rules to Fields

Enter: NSW, VIC, QLD, SA,
WA, TAS, NT, ACT.

Analogy: Like a drop-down
list on a website that
avoids spelling mistakes.



The screenshot shows a 'Lookup Wizard' dialog box. On the left is a preview of a yellow field with a checkmark icon. The main area contains the following text and controls:

- Question: "What label would you like for your lookup field?"
Input: "State" (highlighted)
- Question: "Do you want to limit entries to the choices?"
Control: ☐ Limit To List
- Question: "Do you want to store multiple values for this lookup?"
Control: ☐ Allow Multiple Values
- Text: "Those are all the answers the wizard needs to create your lookup field."

At the bottom are four buttons: "Cancel", "< Back", "Next >", and "Finish".

Activity 1: Building a Table and Adding Rules to Fields

Datasheet View

Week5 : Database- C:\Users\gaming 15\OneDrive - The University Of Newcastle\Documents\

File Home Create External Data Database Tools Help Table Fields Table Tell me what you want to do

Views | Clipboard | Sort & Filter | Records | Find

Save Undo Redo

All Acces... <

Search...

Tables

Table1

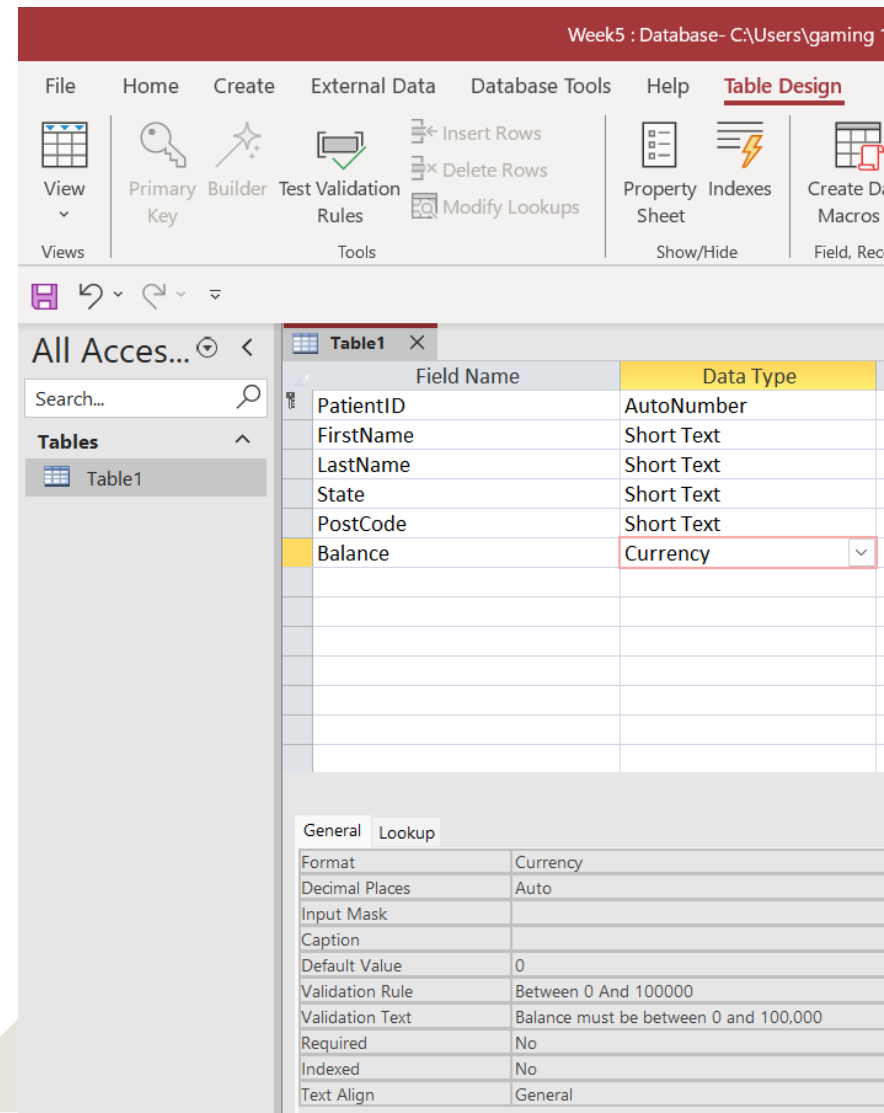
Table1

PatientID	FirstName	LastName	State	PostCode	Balance	Click to Add
1	Farshid	Keivanian	NSW	2000	\$5,000.00	
2	Mary	Brown	NSW	3000	\$12,000.00	
3			VIC			
(New)			QLD		\$0.00	
			SA			
			WA			
			TAS			
			NT			
			ACT			

Activity 1: Building a Table and Adding Rules to Fields

Step 4: Extra Rule – Balance Range

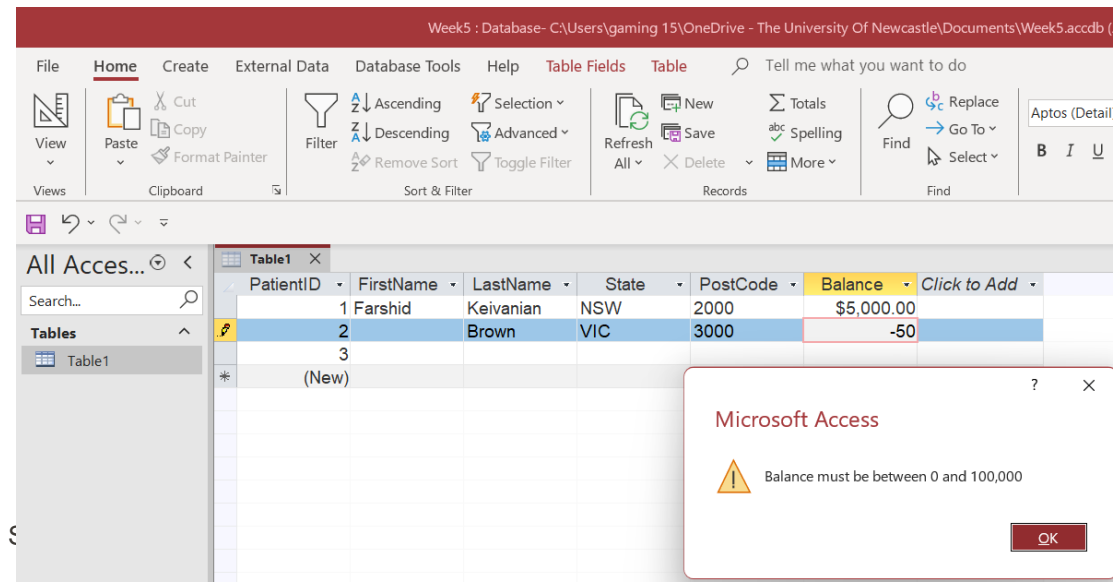
- Still in *Balance*, set:
 - Validation Rule: Between 0 And 100000
 - Validation Text: "Balance must be between 0 and 100,000".



Activity 1: Building a Table and Adding Rules to Fields

Final Hands-on Check

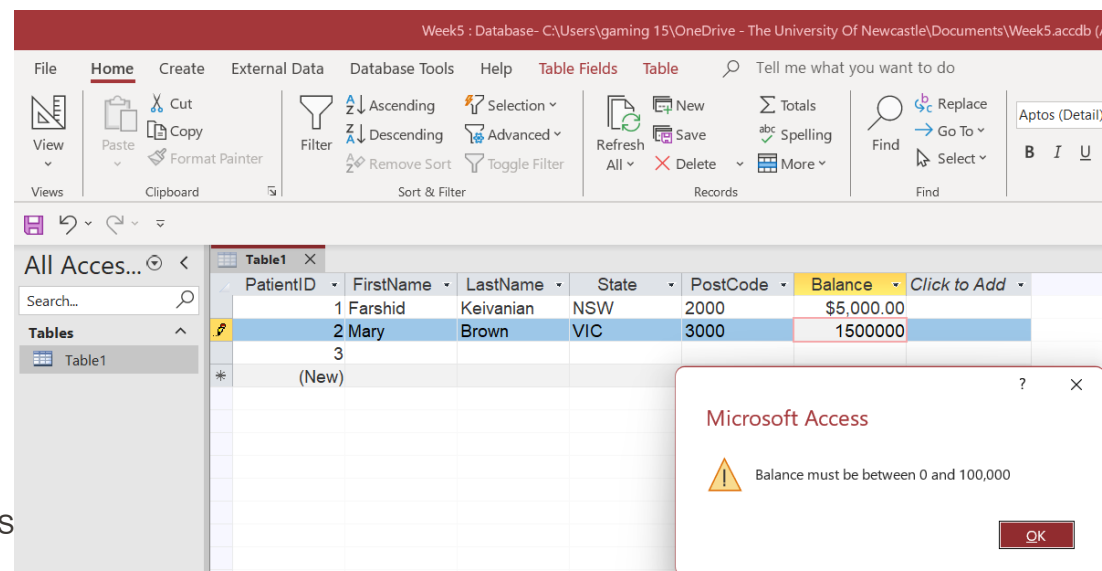
- Try entering a postcode with 5 digits → Access blocks it.
- Try leaving *FirstName* empty → Access forces entry.
- Try entering -50 for Balance → Access blocks it.
- Try entering 150,000 for Balance → Access blocks it.
- State now shows as a dropdown.



Activity 1: Building a Table and Adding Rules to Fields

Final Hands-on Check

- Try entering a postcode with 5 digits → Access blocks it.
- Try leaving *FirstName* empty → Access forces entry.
- Try entering -50 for Balance → Access blocks it.
- Try entering 150,000 for Balance → Access blocks it.
- State now shows as a dropdown.



Activity 2: Mini-System: City & Student

Goal: Create two tables in Microsoft Access, enforce rules, relate them, add data, and test referential integrity.

What you must deliver

1. Table City and table Student built exactly as next slide.
2. Relationship City.CityID (1) $\rightarrow$ Student.CityID (∞) with Enforce Referential Integrity.
3. At least 3 City rows and 5 Student rows linked to those cities.
4. A short note stating whether you would enable Cascade Delete and why.

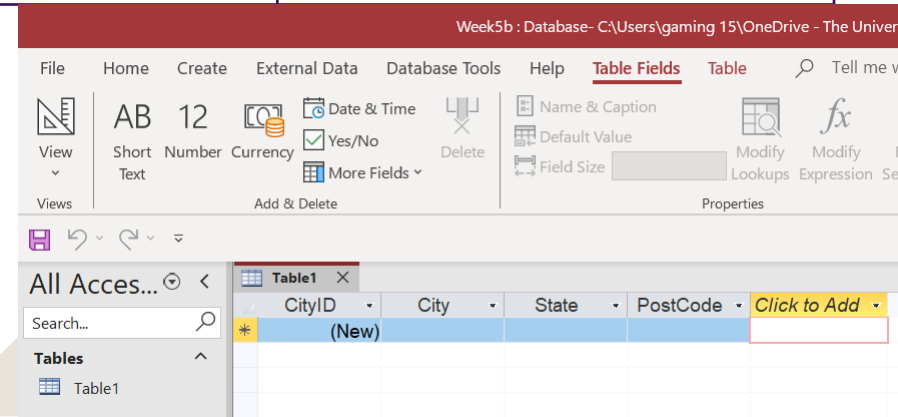
Activity 2: Mini-System: City & Student

1) Create City

Create > Table Design.

Add fields:

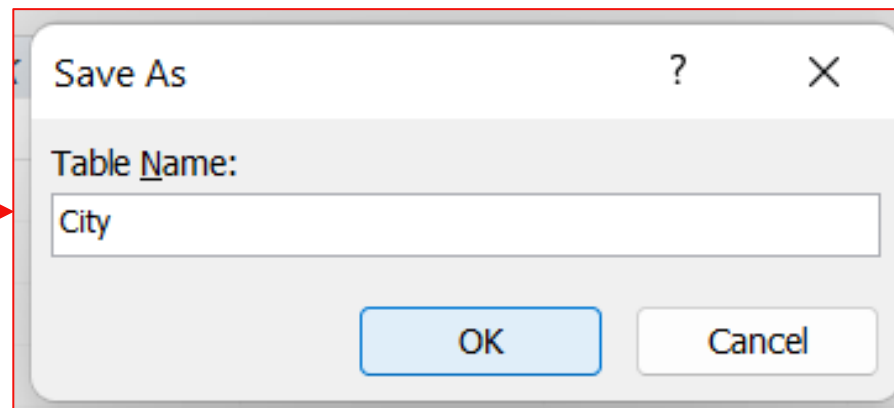
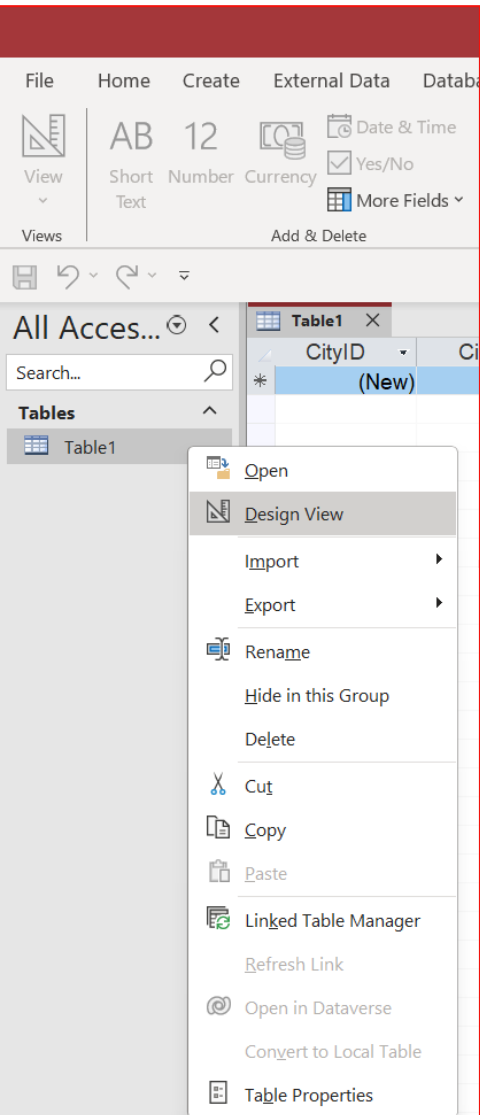
Field Name	Data Type	Properties
CityID	AutoNumber	Primary Key
City	Short Text	Field Size 100
State	Short Text	We'll convert to a Lookup next
PostCode	Short Text	Input Mask = 0000 (four digits)



Activity 2: Mini-System: City & Student

Right Click >> Design View

- Right-click on Table1 in the left panel (Navigation Pane).
- Choose Design View.
- Access will prompt you to save the table → name it City.



Activity 2: Mini-System: City & Student

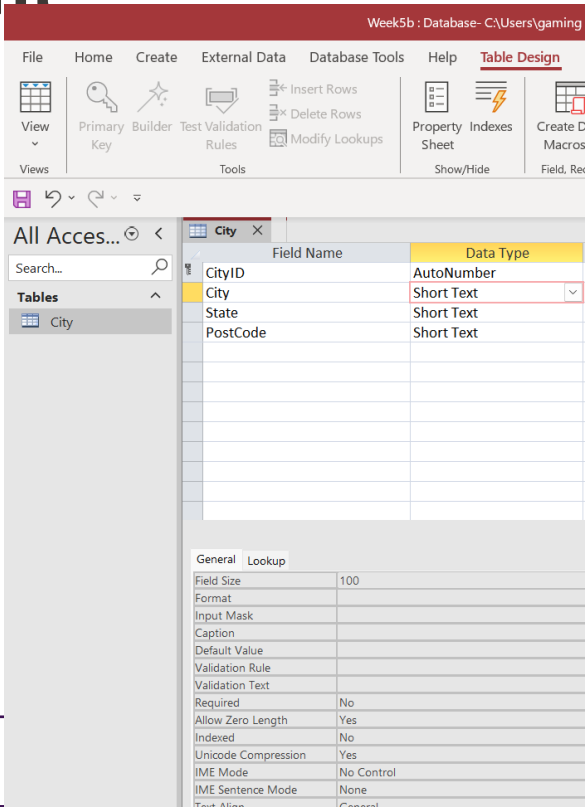
Define Each Field

In Design View, we'll see columns:

Field Name | Data Type | Description

Fill them as follows:

Field Name	Data Type	Properties to Set
CityID	AutoNumber	Set as Primary Key (click the yellow key icon in toolbar).
City	Short Text	In Field Size, type 100.
State	Short Text	Keep as Short Text for now → we'll later convert to Lookup Wizard.
PostCode	Short Text	In Input Mask, set 0000 (forces 4 digits only).



Activity 2: Mini-System: City & Student

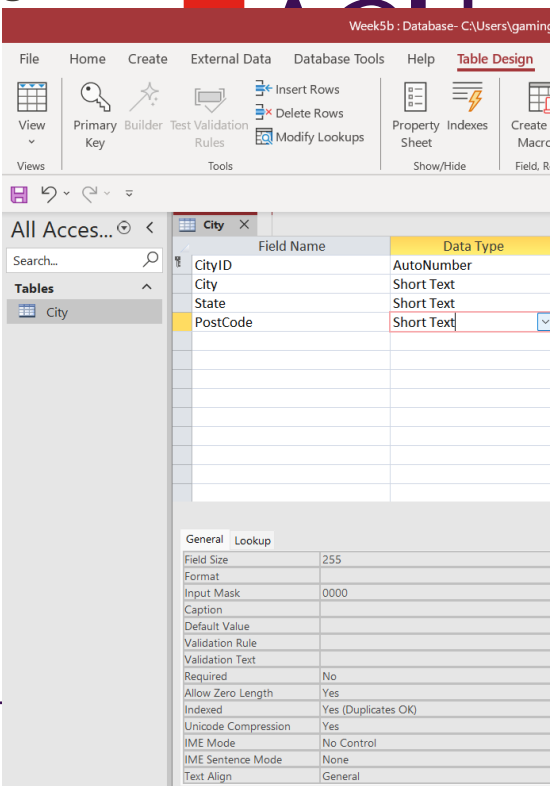
Define Each Field

In Design View, we'll see columns:

Field Name | Data Type | Description

Fill them as follows:

Field Name	Data Type	Properties to Set
CityID	AutoNumber	Set as Primary Key (click the yellow key icon in toolbar).
City	Short Text	In Field Size, type 100.
State	Short Text	Keep as Short Text for now → we'll later convert to Lookup Wizard.
PostCode	Short Text	In Input Mask, set 0000 (forces 4 digits only).

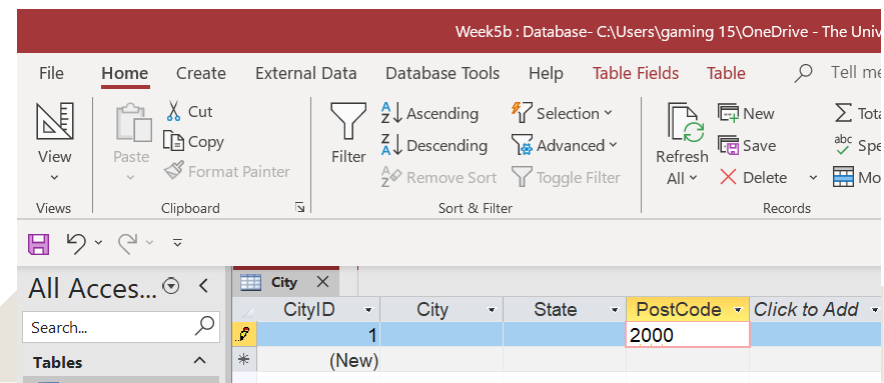


Activity 2: Mini-System: City & Student

Add the Input Mask (PostCode)

- Select PostCode row.
- At the bottom (Field Properties), find Input Mask.
- Click the ... button → Input Mask Wizard opens.
- Choose Zip Code → edit it to 0000.
- Finish → now only 4 digits allowed.

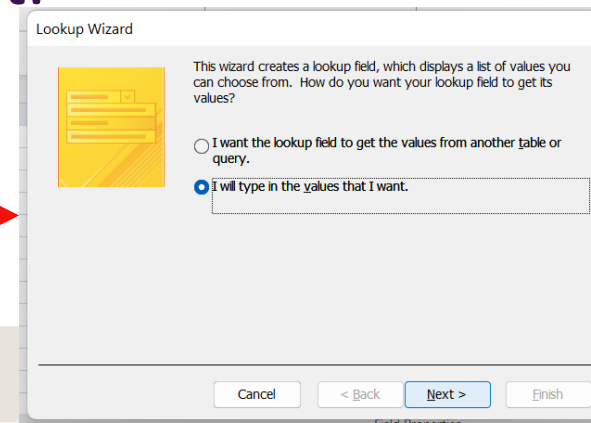
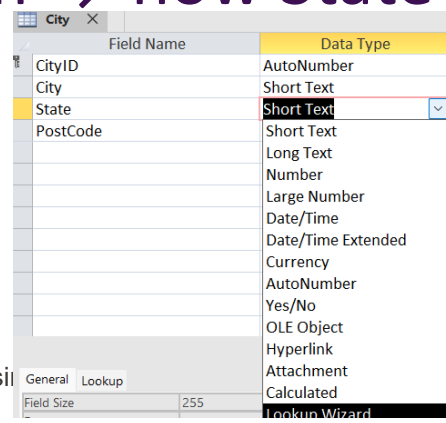
Example: entering 2000 (Sydney) works; entering ABC or 12345 is rejected.



Activity 2: Mini-System: City & Student

Convert State into Lookup

- Select State row.
- Change Data Type from Short Text → Lookup Wizard.
- Choose: *I will type in the values.*
- Enter Australian states: NSW, VIC, QLD, SA, WA, TAS, NT, ACT.
- Finish → now State is a dropdown list.



Activity 2: Mini-System: City & Student

Convert State into Lookup

- Select State row.
- Change Data Type from Short Text → Lookup Wizard.
- Choose: *I will type in the values.*
- Enter Australian states: NSW, VIC, QLD, SA, WA, TAS, NT, ACT.
- Finish → now State is a dropdown list.

Lookup Wizard

What values do you want to see in your lookup field? Enter the number of columns you want in the list, and then type the values you want in each cell.

To adjust the width of a column, drag its right edge to the width you want, or double-click the right edge of the column heading to get the best fit.

Number of columns:

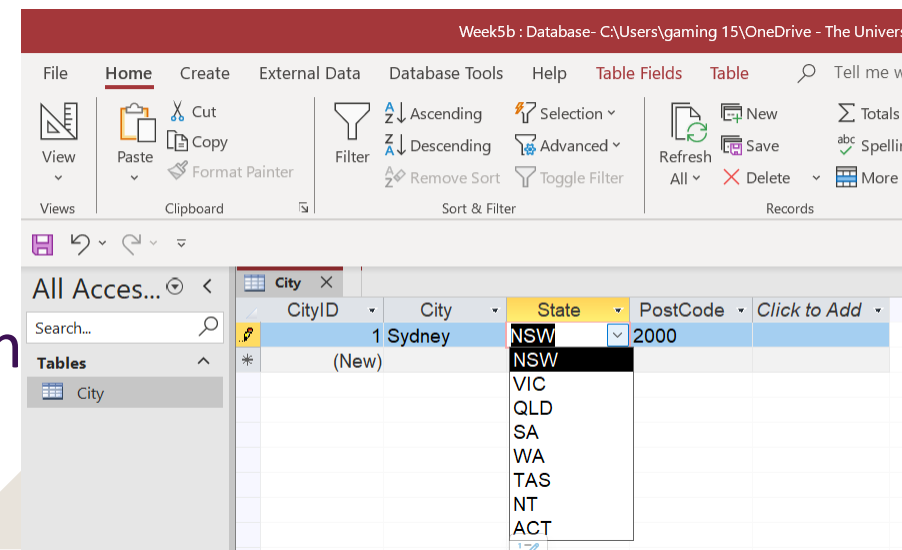
Col1
VIC
QLD
SA
WA
TAS
NT
ACT

Cancel < Back Next > Finish

Activity 2: Mini-System: City & Student

Save & Test

- Save the table (Ctrl+S).
- Switch to Datasheet View.
- Try entering:
 - City = Sydney, State = NSW, PostCode = 2000.
 - If you type 12345 → Access will block it.
 - If you try typing letters in PostCode → Access blocks it.
 - State now shows a dropdown

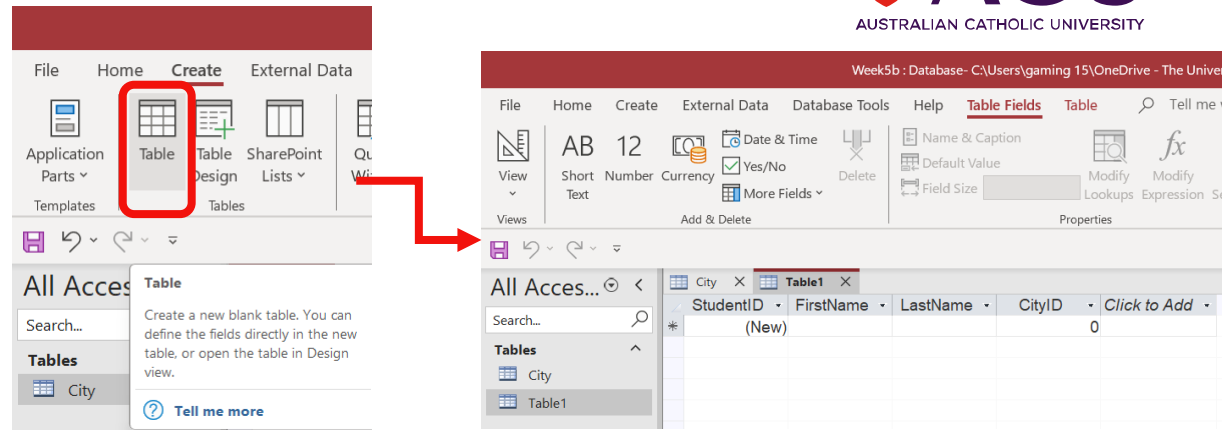


Activity 2: Mini-System: City & Student

Create Student

Create > Table Design.

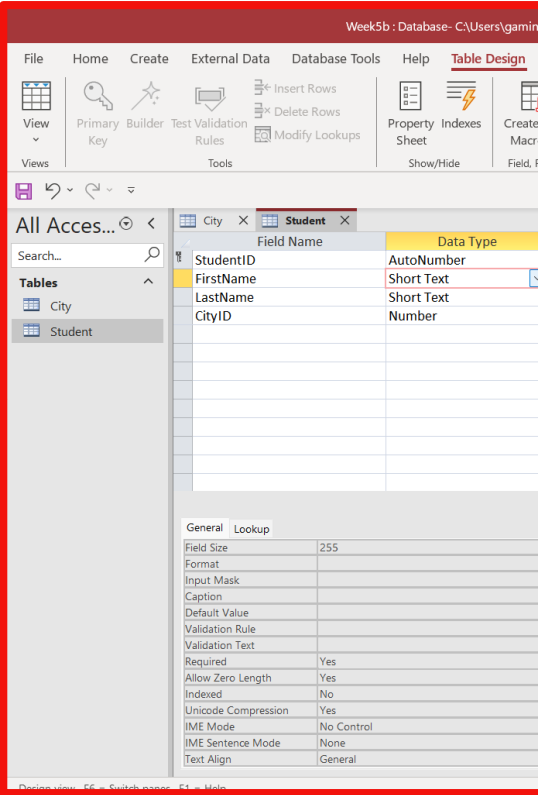
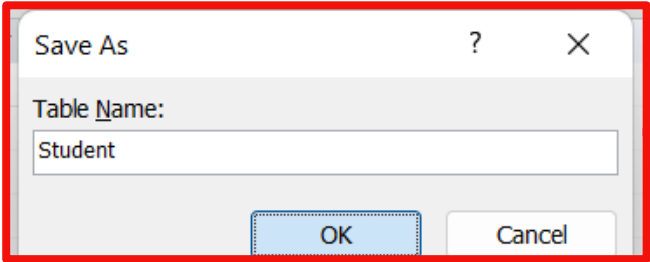
Add fields:



Field Name	Data Type	Properties
StudentID	AutoNumber	Primary Key
FirstName	Short Text	Required = Yes
LastName	Short Text	Required = Yes
CityID	Number	Long Integer (will become a Lookup to City)

Activity 2: Mini-System: City & Student

Click on Design View >> Choose a name



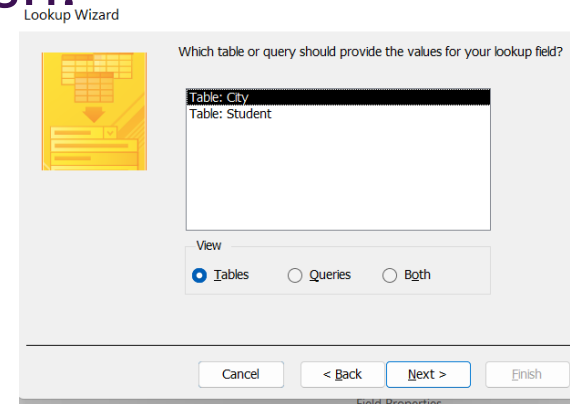
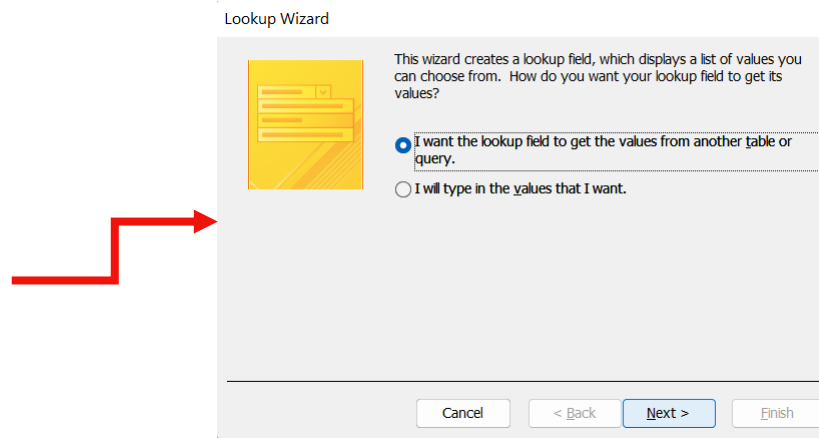
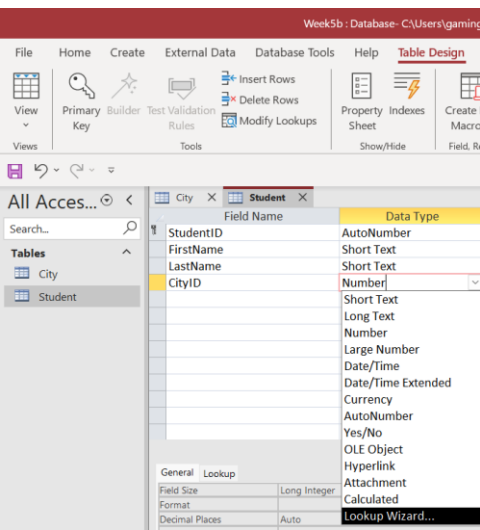
Field Name	Data Type	Properties
StudentID	AutoNumber	Primary Key
FirstName	Short Text	Required = Yes
LastName	Short Text	Required = Yes
CityID	Number	Long Integer (will become a Lookup to City)

Activity 2: Mini-System: City & Student

3. Make CityID a lookup to the City table:

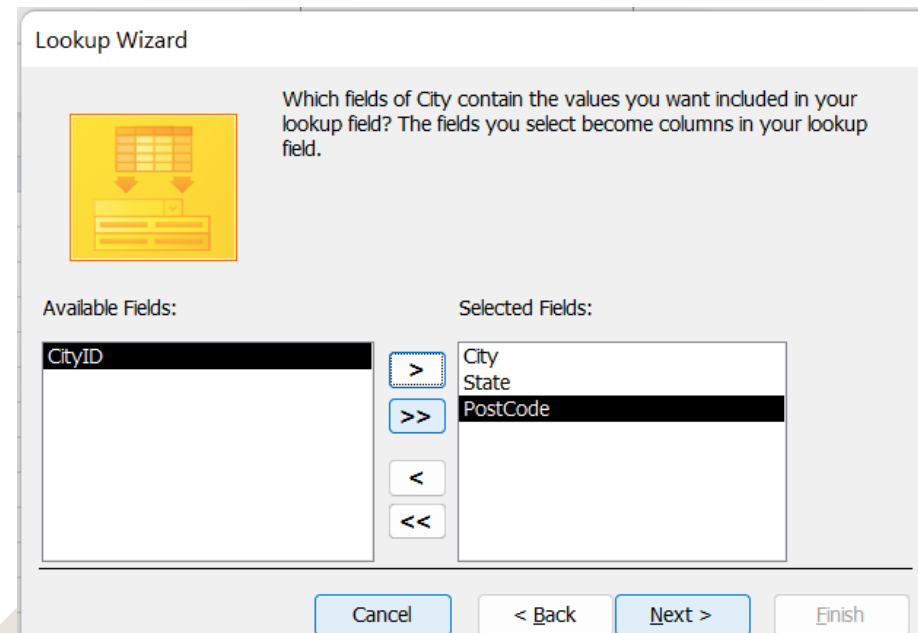
- Select CityID → Lookup Wizard → “I want the lookup field to get the values from another table” → choose City → include columns City, State, PostCode (keep CityID as the bound/hidden key) → sort by City → finish.

4. Save as Student.



Activity 2: Mini-System: City & Student

3. Make CityID a lookup to the City table:
 - Select CityID → Lookup Wizard → “I want the lookup field to get the values from another table” → choose City → include columns City, State, PostCode (keep CityID as the bound/hidden key) → sort by City (so its east to use in the dropdown)→ finish.
4. Save as Student.



Lookup Wizard

Which fields of City contain the values you want included in your lookup field? The fields you select become columns in your lookup field.

Available Fields:

CityID

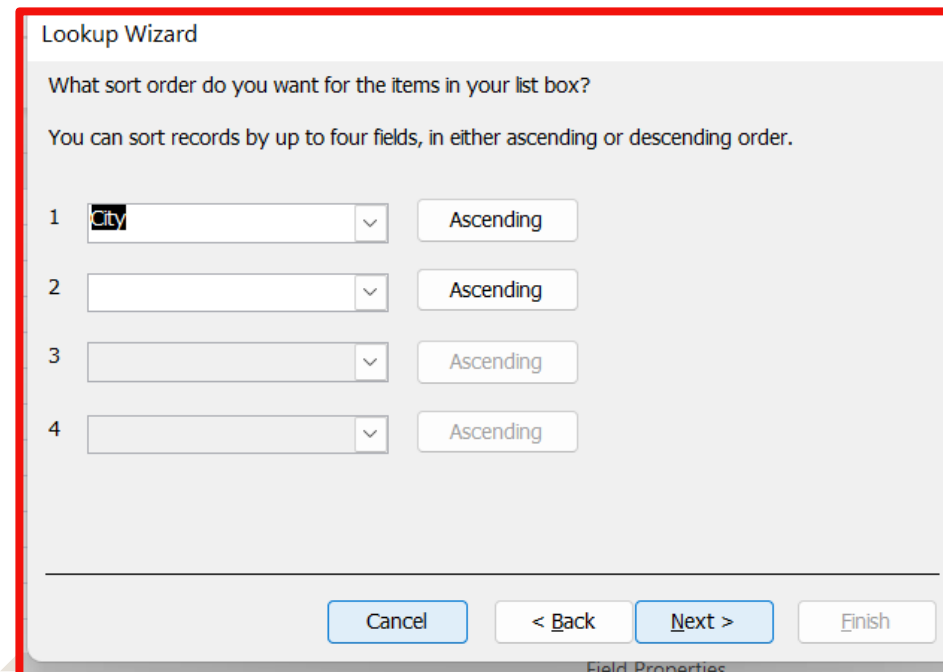
Selected Fields:

City
State
PostCode

Buttons: > >> < << Cancel < Back Next > Finish

Activity 2: Mini-System: City & Student

3. Make CityID a lookup to the City table:
 - Select CityID → Lookup Wizard → “I want the lookup field to get the values from another table” → choose City → include columns City, State, PostCode (keep CityID as the bound/hidden key) → sort by City (so its east to use in the dropdown)→ finish.
4. Save as Student.



Lookup Wizard

What sort order do you want for the items in your list box?

You can sort records by up to four fields, in either ascending or descending order.

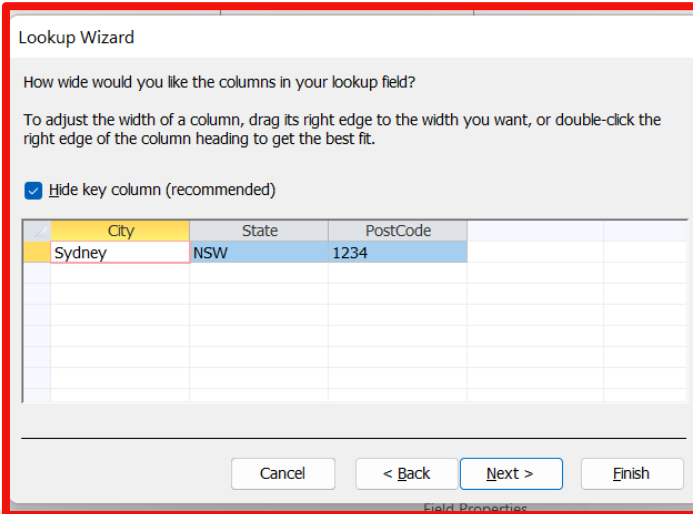
1	City	Ascending
2		Ascending
3		Ascending
4		Ascending

Cancel < Back Next > Finish

Activity 2: Mini-System: City & Student

We'll get the option to *hide the key column*. Tick that, so CityID stays hidden and users only see City/State/PostCode in the dropdown.

- Keep Hide key column (recommended) ticked → that way, the CityID (the numeric PK) stays hidden, but Access will still use it “behind the scenes.”
- The user will instead see *City, State, PostCode* in the dropdown, which is exactly what you want (easy to read, but still linked to the correct ID).



Lookup Wizard

How wide would you like the columns in your lookup field?

To adjust the width of a column, drag its right edge to the width you want, or double-click the right edge of the column heading to get the best fit.

☒ Hide key column (recommended)

City	State	PostCode
Sydney	NSW	1234

Cancel < Back Next > Finish

Activity 2: Mini-System: City & Student

- Click Next → then in the final step, confirm CityID is the bound column.

Then click Finish.

Save the table as Student.

Lookup Wizard



What label would you like for your lookup field?

Do you want to enable data integrity between these tables?

☐ Enable Data Integrity

☐ Cascade Delete

☒ Restrict Delete

Do you want to store multiple values for this lookup?

☐ Allow Multiple Values

Those are all the answers the wizard needs to create your lookup field.

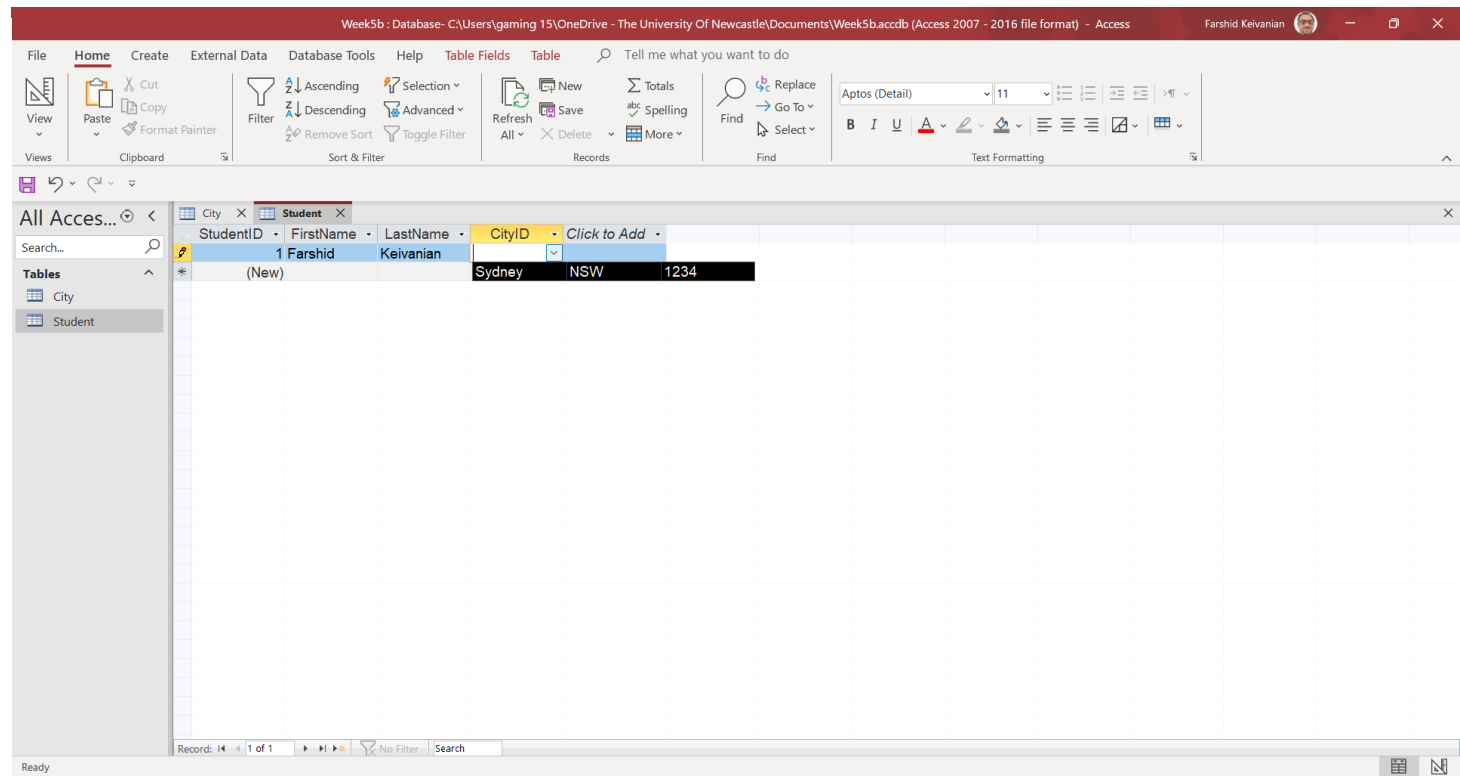
Cancel < Back Next > Finish

Activity 2: Mini-System: City & Student

Test

- In Datasheet View, when you add a Student, the CityID field will now be a dropdown showing City, State, and PostCode.

Underneath,
Access is still
storing the
numeric CityID
(referential
integrity is
preserved).

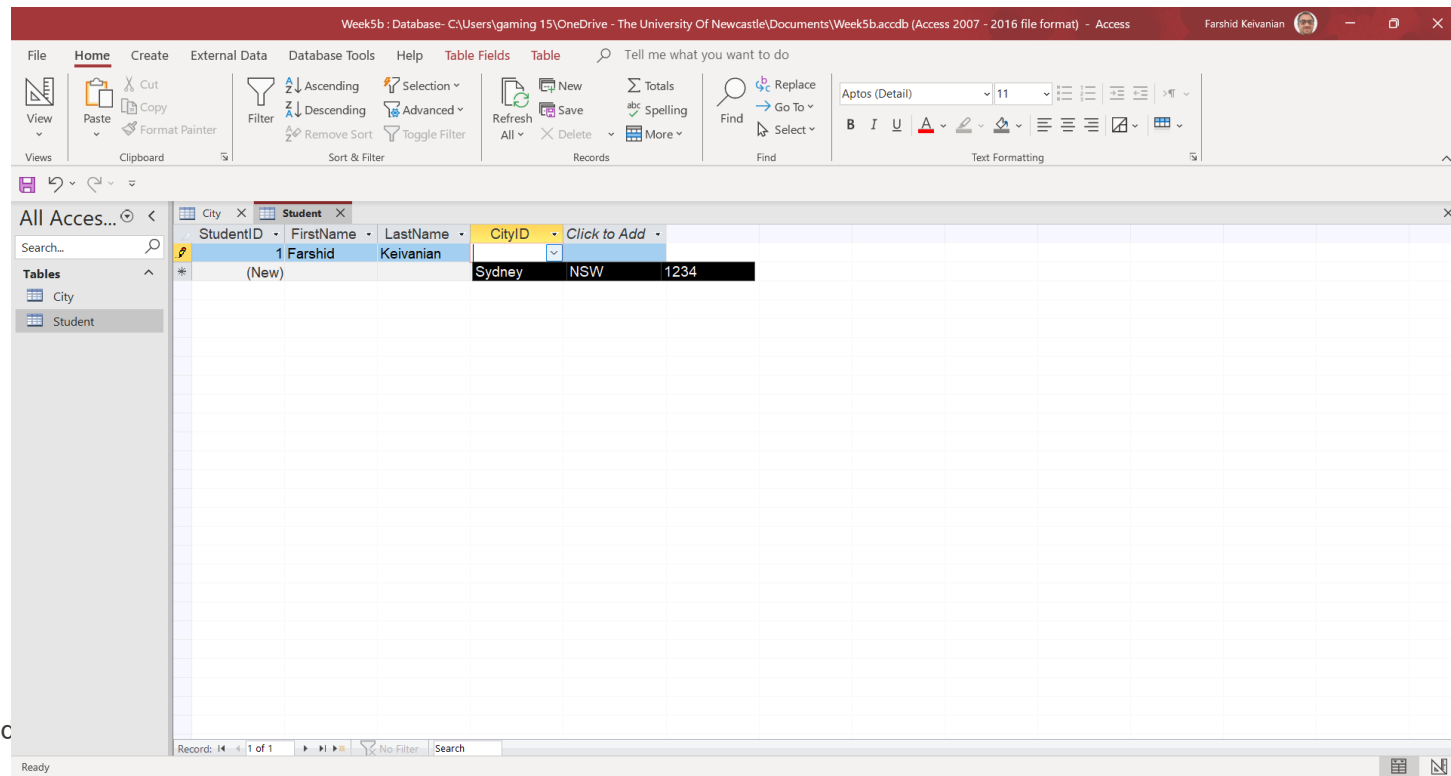


Activity 2: Mini-System: City & Student

It's like filling out a uni enrolment form:

You don't write "CityID = 2" — instead you pick "Brisbane, QLD, 4000" from a dropdown.

The system behind the scenes saves the ID 2, not the text.

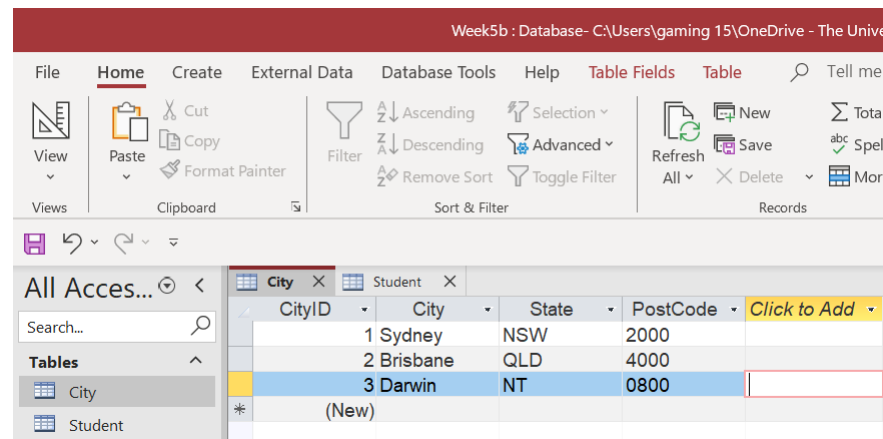


Activity 2: Mini-System: City & Student

Enter sample data

City (add at least 3):

City (example)	State	PostCode
Sydney	NSW	2000
Brisbane	QLD	4000
Darwin	NT	0800

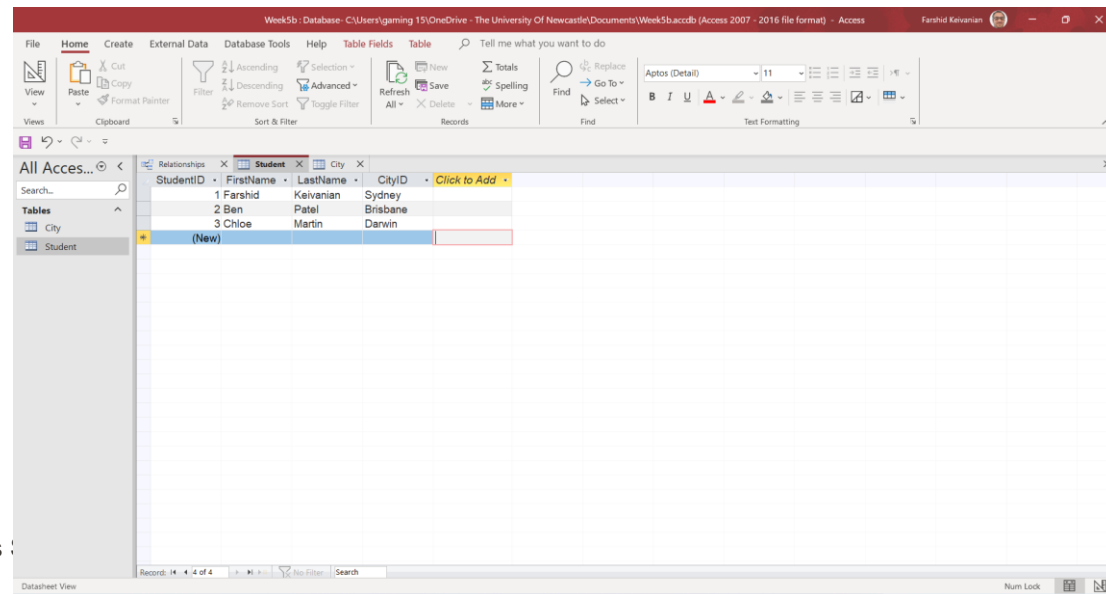


Activity 2: Mini-System: City & Student

Student (add at least 3):

FirstName	LastName	CityID (pick from dropdown)
Farshid	Keivanian	Sydney
Ben	Patel	Brisbane
Chloe	Martin	Darwin

Tip: In Datasheet view we'll see City/State/PostCode in the dropdown, but Access stores the CityID key.



Activity 2: Mini-System: City & Student

4) Create the relationship

Database Tools > Relationships.

Add tables City and Student.

Drag City.CityID → Student.CityID.

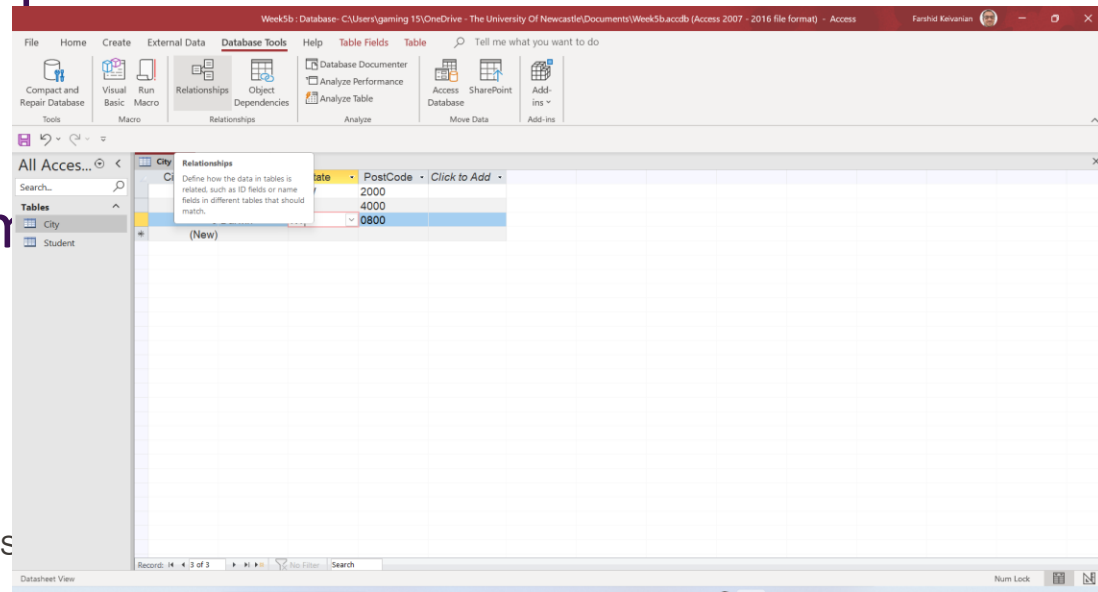
Tick Enforce Referential Integrity.

(Optional) Tick Cascade Update.

Do NOT tick Cascade

Delete for now (safer den

Create, then Save (Ctrl+S).



Activity 2: Mini-System: City & Student

4) Create the relationship

- Database Tools > Relationships.
- Close City and Student Tables
- Then Add tables City and Student.

Drag City.CityID → Student.CityID.

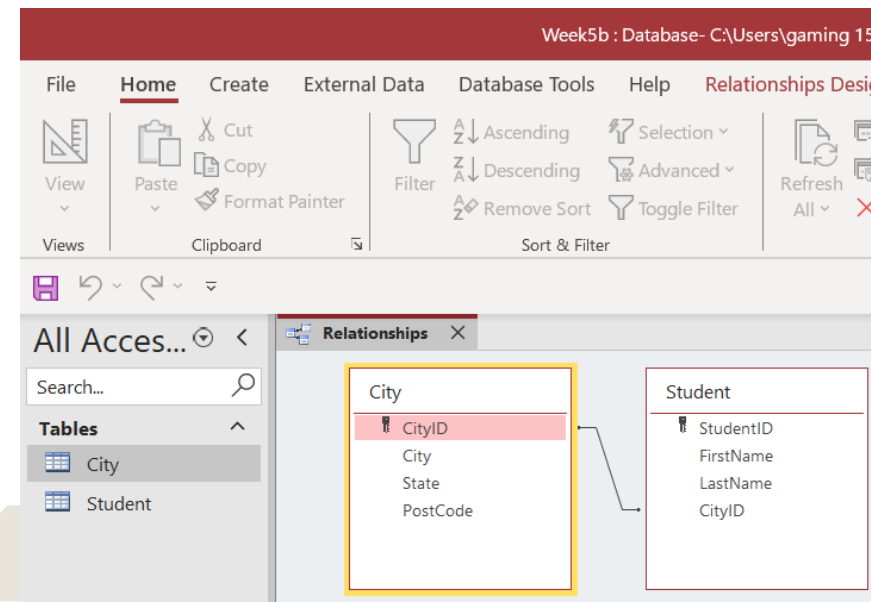
Tick Enforce Referential Integrity.

(Optional) Tick Cascade Update.

Do NOT tick Cascade

Delete for now (safer demo).

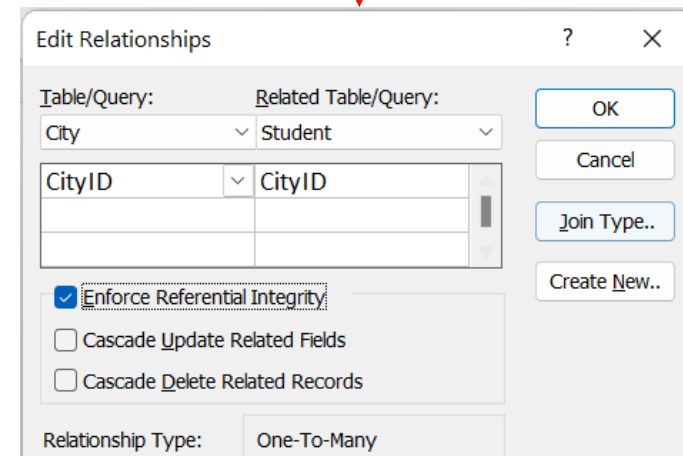
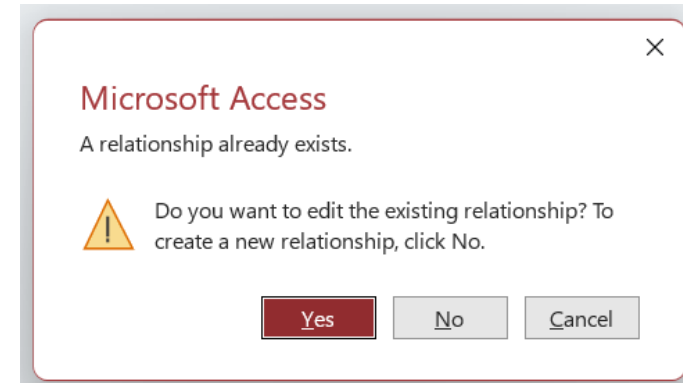
Create, then Save (Ctrl+S).



Activity 2: Mini-System: City & Student

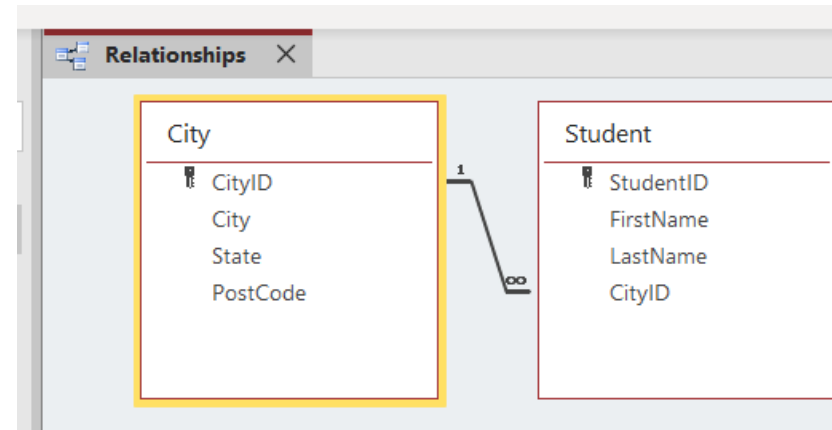
- City → Student
- Primary key: City.CityID
- Foreign key: Student.CityID
- Relationship type: One-to-Many
(one city can have many students).
- Enforce Referential Integrity is ticked.

So it's safe to click OK.



Activity 2: Mini-System: City & Student

- City → Student
 - Primary key: City.CityID
 - Foreign key: Student.CityID
 - Relationship type: One-to-Many
(one city can have many students).
 - Enforce Referential Integrity is ticked.
- So it's safe to click OK.



Activity 3: Library Books & Loans



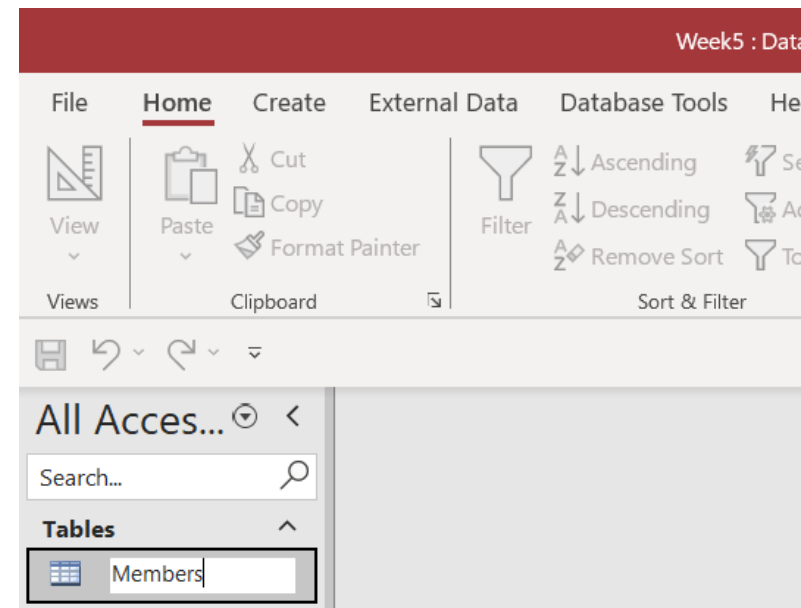
Ctrl + W (Close Table1 from
Activity 1), then rename it to Members

2A – Create & Populate Tables

Goal: Track who has which book.

Tables to Create:

- **Members** → MemberID, FullName, Email
- **Books** → ISBN, Title, Author, Copies
- **Loans** → LoanID, MemberID, ISBN, LoanDate, DueDate, Returned



Activity 3: Library Books & Loans



Ctrl + W (Close Table1 from Activity 1), then rename it to Members

2A – Create & Populate Tables

Goal: Track who has which book.

Tables to Create:

- **Members** → MemberID, FullName, Email
- **Books** → ISBN, Title, Author, Copies
- **Loans** → LoanID, MemberID, ISBN, LoanDate, DueDate, Returned

The screenshot shows the Microsoft Access interface. The title bar indicates the file path: "Week5 : Database- C:\Users\gaming 15\OneDrive - The University Of Newcastle\Documents\Week5". The ribbon includes "File", "Home", "Create", "External Data", "Database Tools", "Help", "Table Fields", and "Table". The "Table Fields" ribbon is active, showing options like "Filter", "Sort & Filter", "Records", and "Find". The "Members" table is open, displaying the following data:

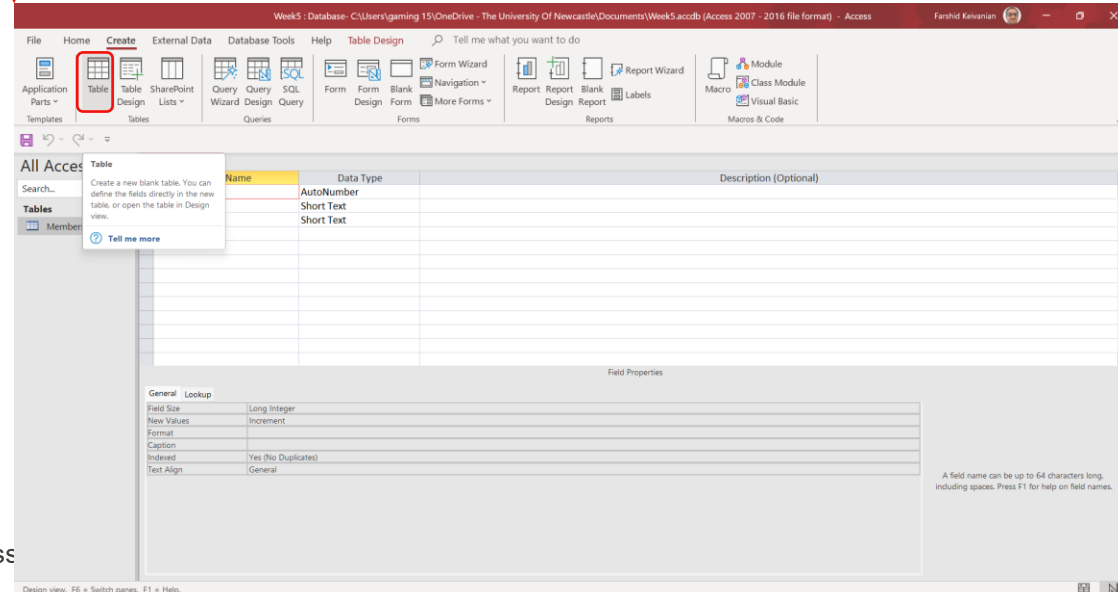
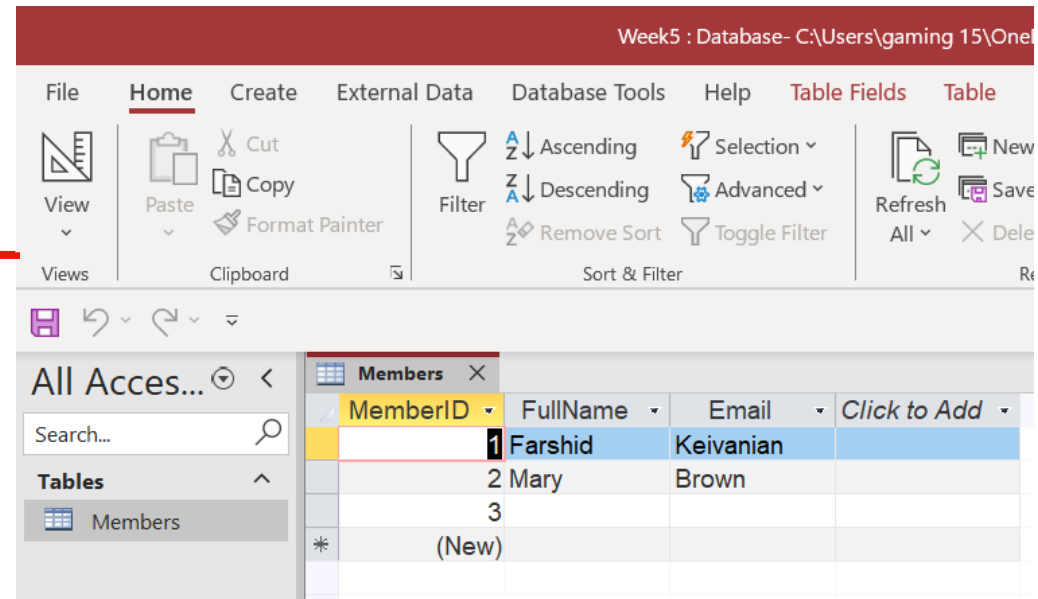
MemberID	FullName	Email	State	PostCode	Balance	Click to Add
1	Farshid	Keivanian	NSW	2000	\$5,000.00	
2	Mary	Brown	VIC	3000	\$15	
3						
*	(New)					

A right-click context menu is open over the table, showing options such as "Sort A to Z", "Sort Z to A", "Copy", "Paste", "Field Width", "Hide Fields", "Unhide Fields", "Freeze Fields", "Unfreeze All Fields", "Find...", "Insert Field", "Modify Lookups", "Modify Expression", and "Rename Field". The "Rename Field" option is highlighted at the bottom of the menu.

Activity 3: Library Books & Loans

Tables to Create:

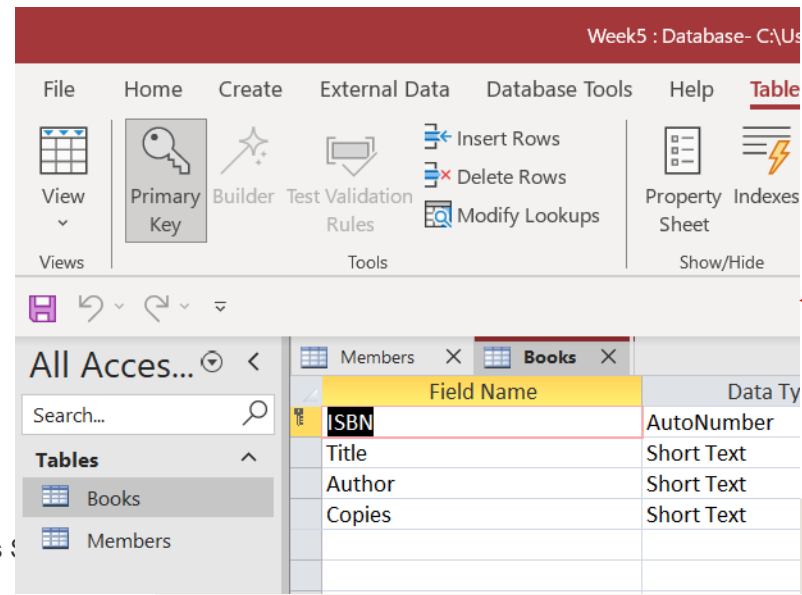
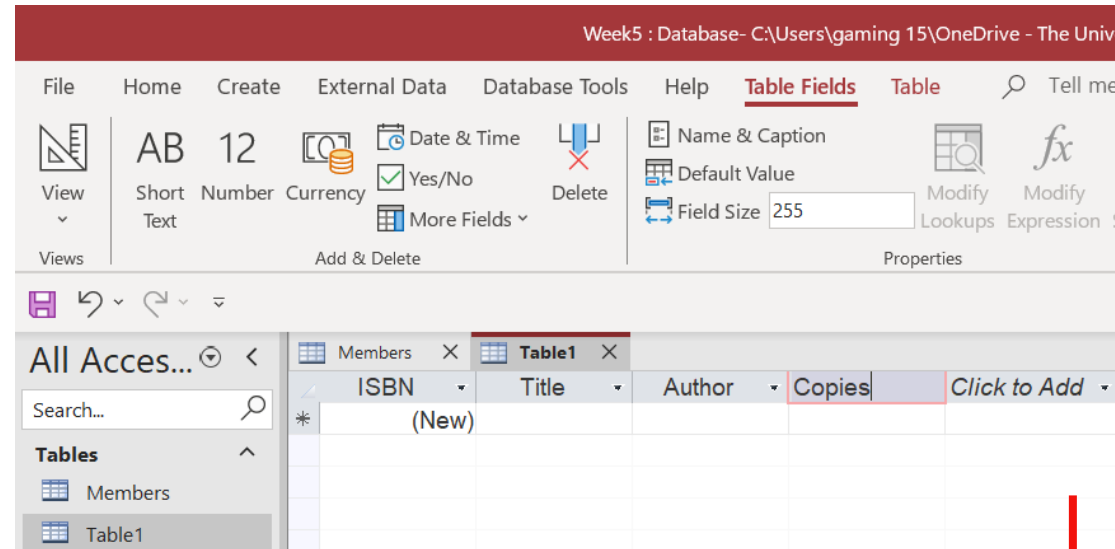
- **Members** → MemberID, FullName, Email
- **Books** → ISBN, Title, Author, Copies
- **Loans** → LoanID, MemberID, ISBN, LoanDate, DueDate, Returned



Activity 3: Library Books & Loans

Tables to Create:

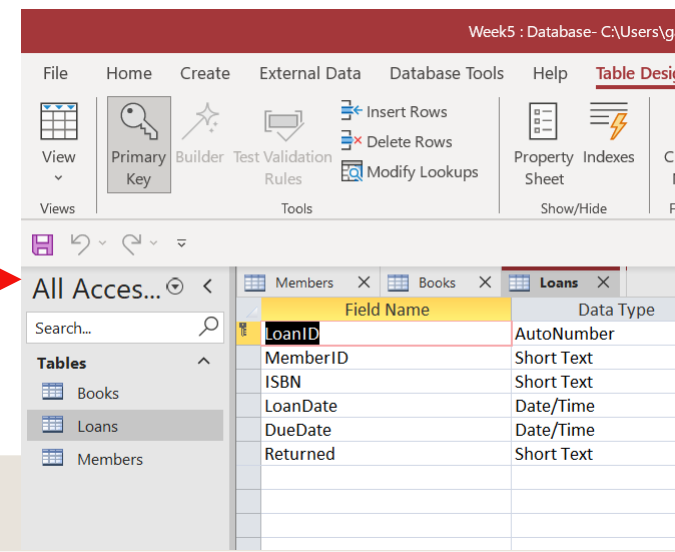
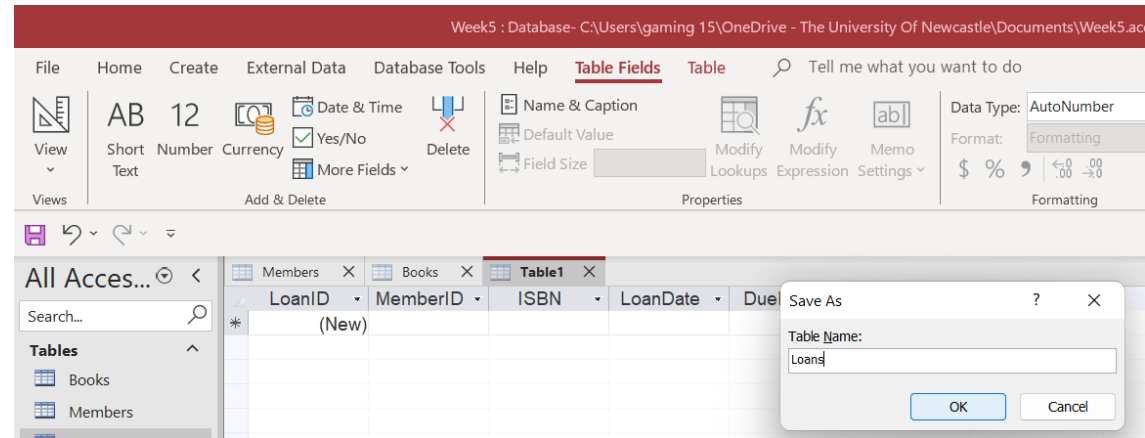
- **Members** → MemberID, FullName, Email
- **Books** → ISBN, Title, Author, Copies
- **Loans** → LoanID, MemberID, ISBN, LoanDate, DueDate, Returned



Activity 3: Library Books & Loans

Tables to Create:

- **Members** → MemberID, FullName, Email
- **Books** → ISBN, Title, Author, Copies
- **Loans** → LoanID, MemberID, ISBN, LoanDate, DueDate, Returned

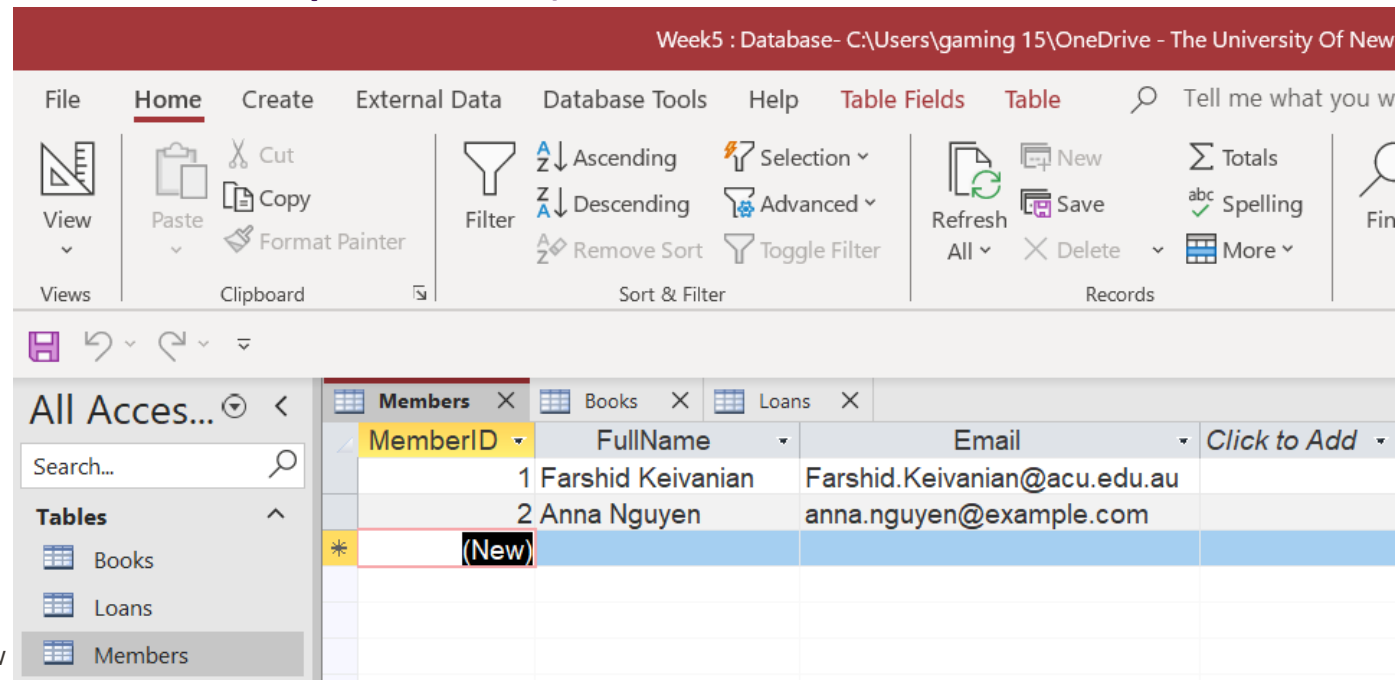


Activity 3: Library Books & Loans

Sample Data (just 1–2 rows each):

- Members: (1, Farshid Keivanian, Farshid.Keivanian@acu.edu.au), (2, Anna Nguyen, anna.nguyen@example.com)

Week5 : Database- C:\Users\gaming 15\OneDrive - The University Of New



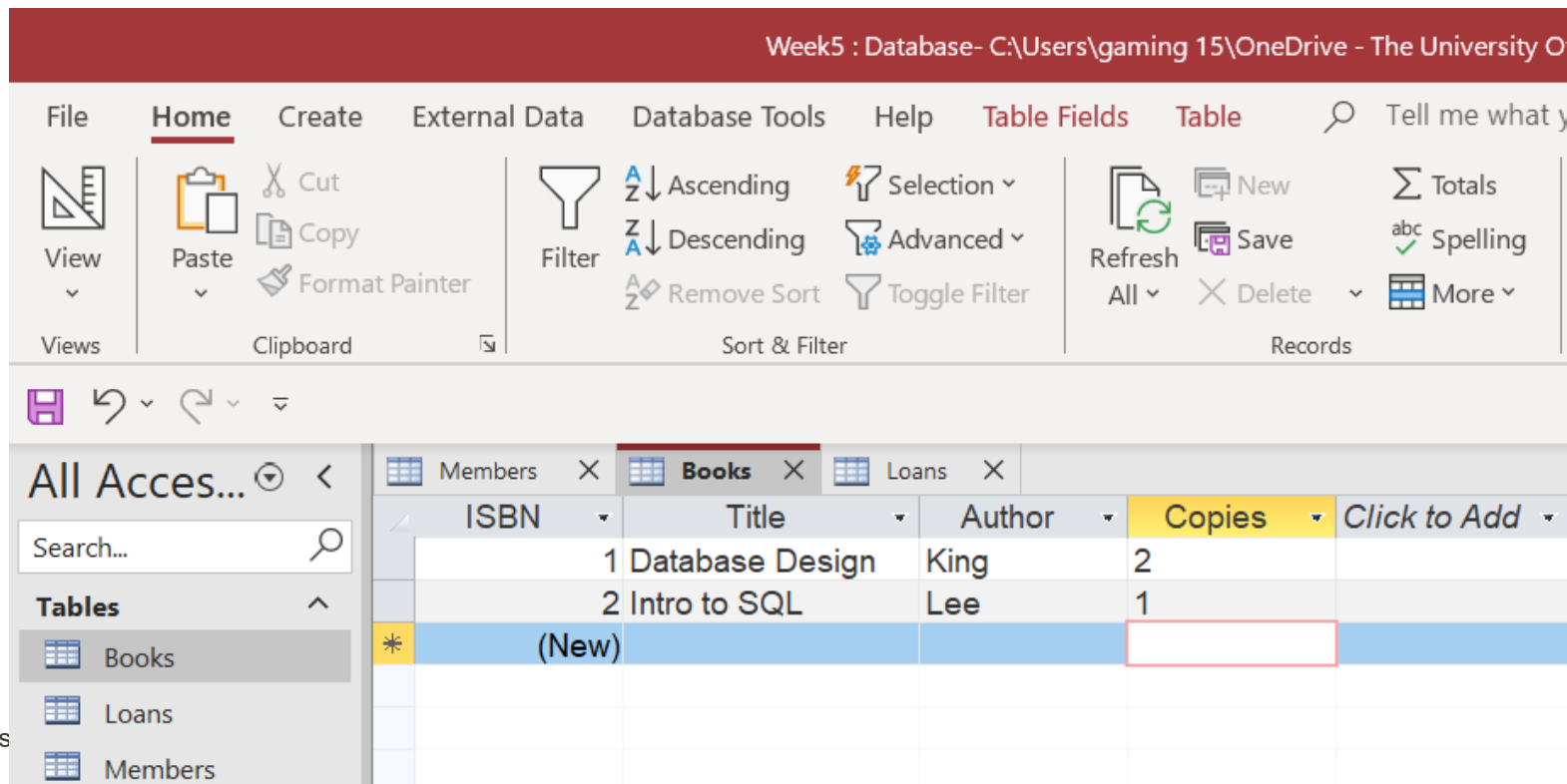
The screenshot displays the Microsoft Access interface. The top ribbon includes tabs for File, Home, Create, External Data, Database Tools, Help, Table Fields, and Table. The Home tab is active, showing options for View, Paste, Cut, Copy, Format Painter, Filter, Sort & Filter (Ascending, Descending, Remove Sort), Selection, Advanced, Toggle Filter, Refresh, New, Save, Delete, and Records (Totals, Spelling, More). The bottom pane shows the 'All Access Objects' list with Tables (Books, Loans, Members) and a search bar. The 'Members' table is selected, and its data is shown in a grid. The grid has columns for MemberID, FullName, Email, and Click to Add. The data rows are: (1, Farshid Keivanian, Farshid.Keivanian@acu.edu.au) and (2, Anna Nguyen, anna.nguyen@example.com). A new row is highlighted with a red border and labeled '(New)'.

MemberID	FullName	Email	Click to Add
1	Farshid Keivanian	Farshid.Keivanian@acu.edu.au	
2	Anna Nguyen	anna.nguyen@example.com	
*	(New)		

Activity 3: Library Books & Loans

- Books: (1, *Database Design*, King, 2), (2, *Intro to SQL*, Lee, 1)
- Loans: (1, 1, 1111, 25/08/2025, 01/09/2025, False)

Week5 : Database- C:\Users\gaming 15\OneDrive - The University O



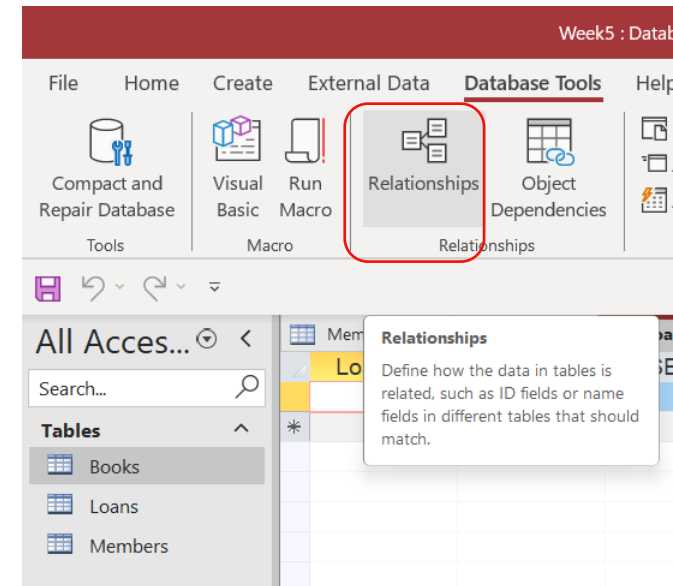
The screenshot shows the Microsoft Access interface. The top ribbon includes tabs for File, Home, Create, External Data, Database Tools, Help, Table Fields, and Table. The Home tab is active, showing options for Views, Clipboard, Sort & Filter, and Records. The Table Fields tab is also visible. The main window displays the 'Books' table with the following data:

ISBN	Title	Author	Copies	Click to Add
1	Database Design	King	2	
2	Intro to SQL	Lee	1	
*	(New)			

The left sidebar shows the 'All Access Objects' pane with a search bar and a list of tables: Books, Loans, and Members. The 'Books' table is currently selected.

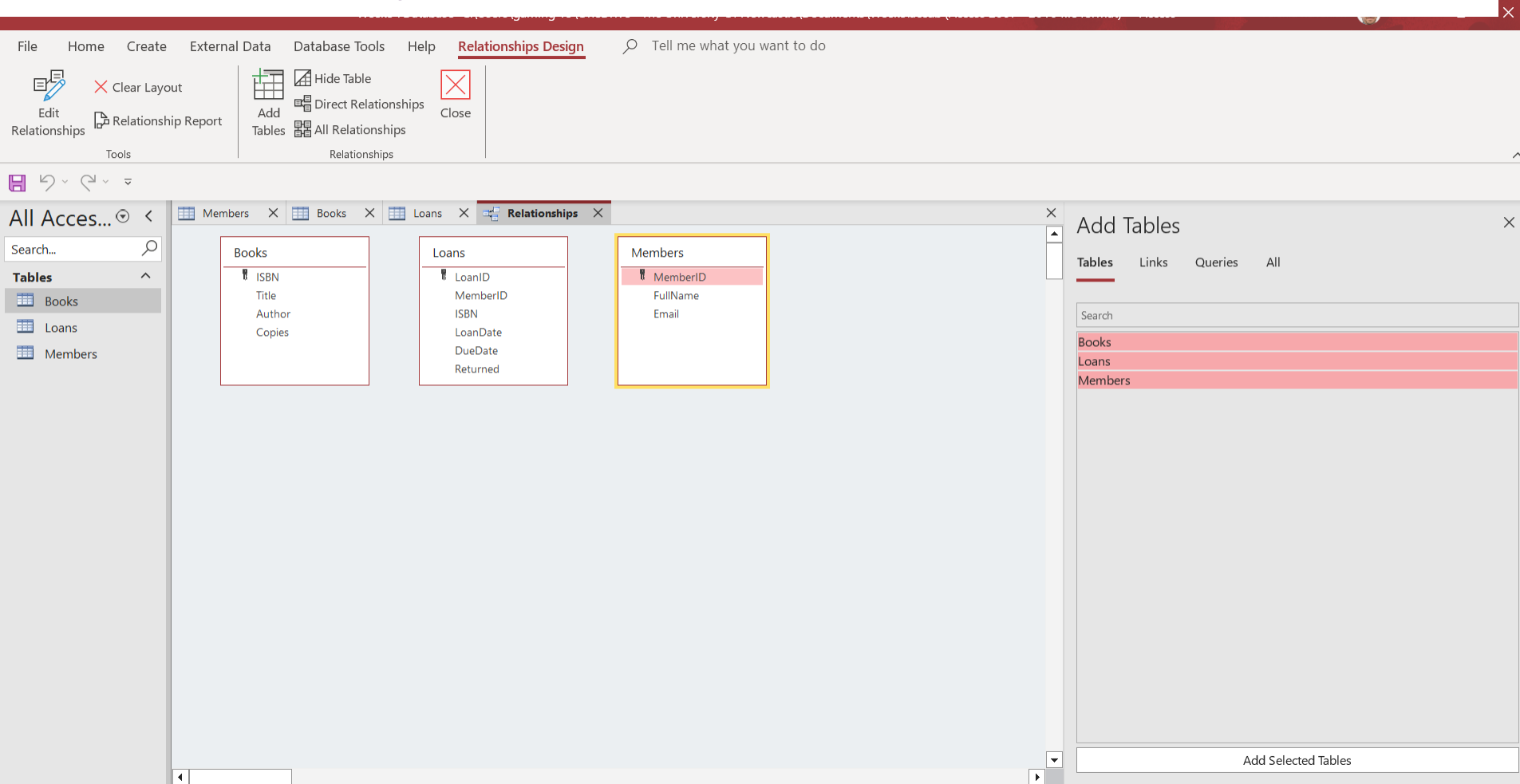
Activity 2B – Define Relationships

- Relationships:
 - Members (1) → Loans (∞)
 - Books (1) → Loans (∞)
- Step: Use Database Tools → Relationships, enforce referential integrity.
- Analogy: One student can borrow many books; one book can be loaned many times.



Activity 3: Library Books & Loans

Close Members, Books and Loans Tables



The screenshot shows the Microsoft Access interface with the **Relationships Design** ribbon selected. The ribbon includes the following groups:

- Tools:** Edit Relationships, Clear Layout, Relationship Report.
- Relationships:** Add Tables, Hide Table, Direct Relationships, All Relationships, Close.

The main workspace displays three table objects: **Books**, **Loans**, and **Members**. The **Members** table is highlighted with a yellow border. The **Books** table fields are ISBN, Title, Author, and Copies. The **Loans** table fields are LoanID, MemberID, ISBN, LoanDate, DueDate, and Returned. The **Members** table fields are MemberID, FullName, and Email.

On the right, the **Add Tables** task pane is open, showing a list of tables: **Books**, **Loans**, and **Members**. The **Tables** tab is selected. At the bottom of the task pane is a button labeled **Add Selected Tables**.

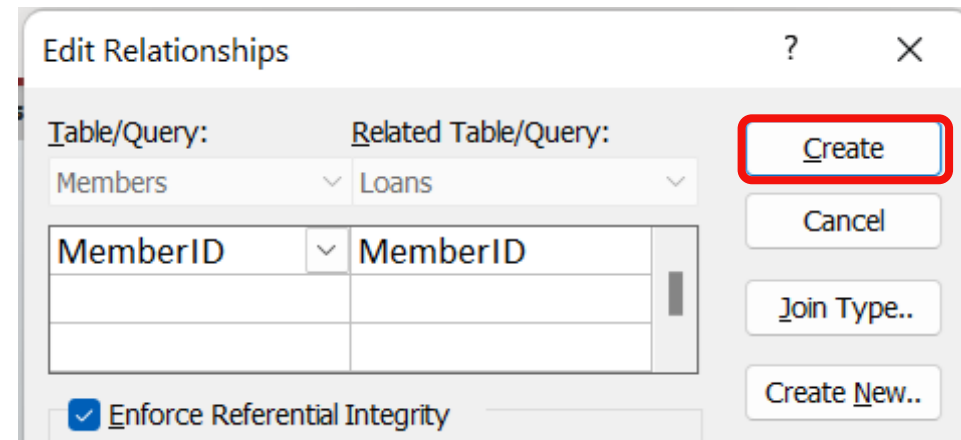
Activity 3: Library Books & Loans



Create the Relationship (Members → Loans)

1. In the Relationships window, drag MemberID from the Members table onto MemberID in the Loans table.
2. A dialog box will appear
3. Enforce Referential Integrity
4. Click Create.

Now we have a line showing
Members (1) → Loans (∞).



Activity 3: Library Books & Loans

Drag ISBN (Books) → ISBN (Loans).

- Members → Loans: One member can have many loans.
- Books → Loans: One book can appear in many loan records.

Activity 3: Library Books & Loans

- **Books → Loans**
 - One ISBN in Books can appear in many rows of Loans.
 - Line shows a 1-to-many relationship.
- **Members → Loans**
 - One MemberID in Members can appear in many rows of Loans.
 - Line shows a 1-to-many relationship.

Activity 3: Library Books & Loans

- One member can borrow many books.
- One book can be loaned many times.
- This enforces referential integrity — no loan record can exist without a valid MemberID and ISBN.

Relationships connect tables (1:M, M:N).

Enforce referential integrity to prevent orphan records.

Example (Australia Health DB):

- One City → Many Patients.
- One Therapist → Many Sessions.

Task 1 Example (City Table):

CityID	City	State	PostCode
1	Sydney	NSW	2000
2	Brisbane	QLD	4000
3	Darwin	NT	0800

Field properties: PostCode = Input mask "0000"; State =
Dropdown list (NSW, VIC, QLD...).

Task 2 Example (Patient Table):

Patient Num	LastName	FirstName	Address	CityID	Balance
1	Nguyen	Anna	12 George St	1	\$250.00
2	Smith	John	55 Queen St	2	\$800.00

Q) Should we always enforce strict validation rules in Access (e.g., PostCode must be 4 digits)?

- **Yes side:** Prevents wrong entries, ensures integrity.
- **No side:** Too strict → may block genuine data entry mistakes.

Answer (wrap-up): Balance is key. Rules improve quality, but flexibility is sometimes needed.



Attendance

Let's Answer Your Questions



- 1. Live during class** – Ask directly during sessions.
- 2. Canvas Inbox** – For structured academic queries.
- 3. Email** – For formal communications or attachments.
- 4. Canvas Discussion Forum** – Everyone can see and learn from your question.

Thank You



Happy Learning and Bright Futures Ahead!

Stay Connected:

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- *Education is the most powerful weapon you can use to change the world. — Nelson Mandela*